

GRUT — Grand Responsive Universe Theory

Consolidated Theory, Formal Architecture, Results, Limits, and Completion Program

D. Ryan Grover

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Abstract

GRUT (Grand Responsive Universe Theory) is a research program. It asks whether local microscopic models can generate continuum, influence, geometric, and gravitationally relevant effective structures without taking those structures as primitive. This record consolidates the program at a deliberate pause. Its central result is a dependency-resolved hierarchy:

$$\mathcal{O}_{\text{local}} \longrightarrow \mathcal{C}_{\text{continuum}} \longrightarrow \mathcal{J} \longrightarrow \mathfrak{A} \longrightarrow \mathcal{G} \longrightarrow \mathcal{F}_{\text{eff}} \longrightarrow \mathcal{G}_{\text{grav}}.$$

The access structure $\mathfrak{A}$ is a cross-cutting condition, not a single link. It fixes which subsystem is retained, which distinctions can be reconstructed, and where boundary events make hidden differences observable.

The core. In the tested classes, eliminating the hidden modes of a finite local network gives the retained part an exact memory kernel. The infinite-volume limit gives continuous spectra and effectively irreversible response. Lattice dimension selects the late-time memory class. These limits are graded as family-level facts, not general theorems. The direction of the arrow of time is supplied, not derived.

Influence. In the declared Gaussian class, realizable response and noise spectra obey $J \geq 0$ and $\nu \geq \hbar J/2$. This is standard Caldeira–Leggett-class realizability, verified in class, with $\hbar$ located as the height of the fluctuation floor. The pair (K, N) is not the whole interface: higher connected cumulants separate environments with identical two-point data. Positivity of the full influence hierarchy is ordinary state positivity, a restatement rather than a new principle.

Access and geometry. A distinction is physical exactly as far as it changes accessible influence data, and the access seed itself is supplied. In the tested classes, rich access reconstructs the hidden network’s geometry, including the elimination of isospectral impostors. Restricted access leaves distinct geometries indistinguishable. Geometry here is access-relative, never absolute.

Nonstationarity. The core construction extends, in class, to finite networks whose couplings change in steps. The reduction is exact to 10^{-12} and continuous with the stationary case. A separately pre-registered test certifies that no single spectral measure and no function of the time difference alone packages the resulting two-time memory. The first test’s two packaging gates failed and remain permanently red.

Gravity. A route toward a gravitational identification was attacked one assumption at a time. Still supplied are the access seed and a coincidence of access seeds (CARRIER), a massless gauge- and Lorentz-redundant probe, universal reach, a co-stretch (unit-change) declaration, and a cross-sector light cone. Given these inputs, the retained sector is constrained to a gapless, $z = 1$, locally accessible Goldstone-like class. A matter-side dissipation exponent, ω^7 , is within-class occupancy evidence only. Its record includes an analytic calculation sealed in advance, but also unimplemented sealed gates and pre-registrations committed together with the result (Section 17). The forcing (“Class-4”) derivation of gravity is open.

What is claimed. No confirmed prediction is claimed. GRUT-specific derived predictions stand at zero, the program’s distinctive-theory adjudication failed as scoped, and seventeen failed pre-registered gates (ten red-register entries) are preserved. No outside human has reviewed the work. The achievement is narrower and more exact. Repeated counterexample-driven attacks have organized a diffuse set of assumptions into an explicit inventory of eleven named primitive inputs, unresolved seams, and preserved red gates. Completion is defined as seven typed mathematical problems, and an input may legitimately remain primitive once its status is certified.

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Source boundary, stated fully.

- *What it covers.* This manuscript consolidates the GRUT record on `master-w25bu9` through commit `6abbbf316`. It also relies on the foundations record on `adjudicator-track` (tip `90218f5`), which this release archives byte-exact because the branch is not merged (Section 28).
- *The RRP line.* `master-w25bu9` forks from the repository’s `master` branch at commit `a2dcb02` (24 September 2026), the RRP program synthesis. The RRP records are therefore in the source history. Their results enter this theory only through deliberate source inspection, scope classification, and owner adjudication.
- *Other branches.* The public repository has other branches (`main`, `physics-final`, `v4`, `v2`, `v1-retired`, `testingi-rrt`, `rrt0-phase2`). They are not sources for this record except where cited by commit.
- *This file’s own commits.* The manuscript and its release tooling were committed after the source boundary. They add no computation and change no status.

Epistemic boundary. This is an in-house research record: computations, formalizations, and owner rulings, each at its declared scope. No external peer review is claimed, and no experimental validation is claimed. No outside human reviewer has checked any GRUT-specific result.

Notice: correction and supersession of an earlier deposit

This version is deposited under the same concept DOI (10.5281/zenodo.19803663) as earlier GRUT work. Among those earlier versions is *Grand Responsive Universe Theory GRUT ToE v4.0 — The Emergence of Everything — Candidate Framework* (version DOI 10.5281/zenodo.20783057, deposited 21 June 2026). **The claims of that deposit are withdrawn. It is substantially**

wrong, and nothing in it should be cited as a result. Its author’s own supersession notice gives the record:

- A later verification pass scored it 1 of 18 on its verifiable numbers, 0 of 7 on its falsifiers, and 1 of 11 on its scorecard rows.
- Several of its lines did not survive:
 - its dark-matter substrate line;
 - its two-anchor hierarchy ledger;
 - its claimed unified no-go;
 - its entire “forbidden-by-theorem” tier.
- Two of its headline claims fail outright:
 - Its advertised 32σ exclusion recomputes to about 2.0σ , and the mechanism it rested on had the wrong sign.
 - Its “689 Hz, zero free parameters” falsifier does not exist as advertised.
- Its title carried the phrase “Theory of Everything”. This record does not.

The later *Program Record* release of 7 September 2026 (Books I–X, working edition) retired the theory-of-everything claim. The de Sitter absorptive-response paper is the previous version of this record and remains a standalone standard-QFT result (Section 24.1). Where any earlier version in this lineage conflicts with the present record, the present record’s scoped statements govern. Section 29 gives the full release history.

What this is and what it found

GRUT began as a hypothesis about gravity. The hypothesis was that the gravitational vacuum behaves like a responsive material with memory: empty space responds to what happens in it with a delay, rather than instantly. That idea gives the program its name.

Its strongest form, a vacuum memory with a single characteristic time, did not survive: exact de Sitter free-field calculations produce memory with no single time scale. In September 2026 the program’s own adjudication found that the framework, as then scoped, “fails to constitute a distinctive physical theory”, which is not a finding that its hypothesis is false. The program was closed on 23 September and reopened on 25 September with a narrower question.

This record answers that narrower question: **in explicit model systems, what structure appears on its own, and what must be put in by hand?** The models are mostly networks of coupled oscillators (masses and springs, or their overdamped analogues) plus small quantum spin systems. Each is only partly observed: some degrees of freedom are watched, and the rest act as a hidden environment.

Found, within the tested model classes only:

- **Memory and irreversibility from local dynamics.** Hiding part of a local network gives the visible part an exact memory. As the network grows, its response becomes effectively irreversible, and the lattice dimension sets the long-time form of the memory. The direction of time is not derived.
- **Averages and noise are not the whole story.** Higher-order correlations can distinguish environments that agree on both the average response and the noise.
- **Access decides what can be learned.** With rich access to the hidden network, its dimension and distances can be reconstructed. From a single site, genuinely different networks look identical.

- **Changing networks need two-time memory.** When the network changes in steps, the memory depends on two times, not only on their difference, and no single-time description packages it. The smooth and infinite-size cases are open.

Not found:

- No experimentally confirmed prediction. The program’s own ledger of GRUT-specific derived predictions stands at zero.
- Planck’s constant, the Born rule, and quantum noncommutativity are inputs, not results.
- Gravity is not derived. A route toward a graviton-like response works only if several structural inputs are supplied by hand. Among them are a massless graviton-like field that reaches every sector, and a light cone shared between sectors. Several pre-registered tests along that route failed and remain on the record.

How the work was done. Much of the mathematics is standard open-systems and spectral theory. The program’s contribution is pre-registered, in-house verification within these classes, and a precise map of what depends on what. The computations were run largely by AI agents under the author’s direction, and the author ruled on each status. No outside human has reviewed any GRUT-specific result.

This record is not a completed theory, and it is not a theory of everything.

How to read this record

- **New to GRUT:** read “What this is and what it found”, Section 0, Section 26, and the ledger of Section 31.
- **Critical specialist:** read Section 32, Sections 4–10, Sections 20–24 and 31, Sections 12–19, Sections 26 and 29, then Section 28 and Appendices D and G.
- **Section 32** gives the strongest supported statement, and **Section 31** the full status ledger.
- **Sections 0–3** cover motivation, method, what GRUT claims, its relation to existing work, and the formal architecture. **Figure 1** (Section 3) is the map of the whole record.
- **Sections 4–11** present the generative core: continuum, stationarity, influence, access, geometry, units, and quantum structure.
- **Sections 12–19** cover the conditional gravitational branch.
- **Sections 20–25** give the formal statement, the input inventory, the seams, the red register, the completion program, and falsifiability.
- **Sections 26–29** give the non-claims, provenance, and history.
- **Appendix H** is a glossary of every program-specific term. **Appendix I** resolves symbols that carry more than one meaning. **Appendix G** indexes every campaign fork by commit.

Each result is graded by a fixed vocabulary (Section 1). Numbers are given with their source, and the repository artifacts decide the status of any individual claim.

Two words used narrowly. In this record, *prediction* means an empirical prediction about nature; agreements between in-house analytic and numerical computations are called checks. *Theory* means the program’s current formal construction (Sections 20–24); the program’s own distinctive-theory adjudication failed as scoped (Section 29), and nothing since has been adjudicated to overturn that grading.

0. Origin, and why this record is narrower

The founding hypothesis. GRUT began as the hypothesis that the gravitational vacuum is a responsive medium with finite memory. In an open-system (in-in) description, its retarded response and noise kernels would carry a single finite memory time τ_0 : a single-pole kernel. The idea was that this would make gravity a universal, memory-bearing environment to which all matter couples. The program’s working identity followed from it: the *gravitational identification*, the claim that the persistent continuum generated by local dynamics is physically realized by gravity. That identification is graded **hypothesis**, not derived (Section 12).

What happened to it.

- The single-pole bet is negated in its strongest form: exact de Sitter free-field theory forces scale-free, power-law memory at the computed order.
- A distinctive-theory adjudication, run twice on 4–5 September 2026, found that the framework “fails to constitute a distinctive physical theory”. The adjudication records that this is **not** “the responsive-vacuum hypothesis is false”. GRUT was left unevaluated at its own claim point.
- On 23 September 2026 the program was formally closed. On 25 September 2026 it was reopened by owner ruling as a theory-development effort, with a narrower object: *exhibit the mathematics of the theory — show how it works — before any external claim is made*.

Reopening is not rehabilitation. The reopening ruling states that every earlier grading stands. That includes zero derived predictions, the failed distinctive-theory adjudication, the negated founding bet, and the no-go ledger.

What this record does. It reports the dependency campaign run after the reopening:

- twenty-two chartered forks;
- a formalization;
- a completion attack on the stationarity seam (C1-a and C1-a2).

It asks what structure remains once every assumption the program could not earn has been removed. It is published at a deliberate consolidation pause. Later results enter only through a new dated record.

1. Scope, epistemic discipline, and status vocabulary

The record is maintained under a source-first rule: **computational checks outrank narrative prose**. A later document describing a claim more confidently does not make the claim stronger.

The formal grading vocabulary (Formalization 01 §0; binding):

- **[IDENTITY]**: an exact mathematical identity. It carries content only as a classification.
- **[DERIVED-IN-CLASS]**: established by a chartered instrument within a declared model class, at stated tolerances. The class is part of the statement.
- **[FAMILY-FACT]**: demonstrated across a tested family of models. It is stronger than an example but is not a general theorem.
- **[SUPPLIED]**: a primitive input the program does not derive (Section 21).
- **[UNDEFINED]**: a composition or object the theory does not currently define. The seams live here.

Other status labels used in the record:

- **[CONSTRAINED]**: restricted by the evidence but not uniquely selected.
- **[DOMAIN DATUM]**: state, boundary, or regime information explicitly typed as data within the theory’s domain. It is one of the three certified closure statuses (Section 24).
- **[UNRESOLVED] / OPEN**: not decided by present evidence.
- **[RED]**: a pre-registered gate failed. **The gate stays red permanently.** A separately chartered measurement may establish a new result on the same question, but it never turns the original gate green (Appendix C).
- **[NULL-AS-NEW-PRINCIPLE]**: a proposed foundational principle that reduced to an existing identity or positivity condition, under the program’s redundancy rule.
- **Owner ruling**: a project-level adjudication by the author recording how a result is admitted, demoted, fenced, or incorporated. It is not a theorem, a peer-review finding, or an experimental confirmation.

A failure is a result. It is never repaired by changing the interpretation after the run.

A conditional result stays conditional. A successful calculation does not discharge the assumptions that made it possible. Program-specific terms are defined in **Appendix H**, and the meaning of “pre-registered” in this record is stated precisely in **Appendix D**.

2. What GRUT currently claims

Within the admitted model classes, local dynamics generate continuum spectral structure, memory, effective dissipation, and an influence hierarchy that is complete as an interface within the declared model and access class. Declared access determines which distinctions are recoverable. In the tested classes, sufficiently rich access reconstructs graph and substrate geometry from accessible influence data. The program has not established a unique microscopic ontology, nor shown that every physical sector must instantiate this construction.

The campaign forced three demotions relative to earlier formulations:

1. **The constitutive kernel is not fundamental.** A memory kernel is downstream of subsystem selection, state information, and access (P-1, D-1). The influence hierarchy is also richer than the kernel pair (K, N) (P-3).
2. **Stationarity is not fundamental.** Exact nonstationary constructions produce genuinely two-time kernels (Section 5). The stationary form $K(t - t')$ is a regime-specific representation.
3. **In the tested class, “medium” versus “relational” language is not physical.** In the exactly solvable Gaussian class, with a factorized initial state, genuinely different microscopic environments are observationally equivalent at fixed influence data and system-limited access (D-1). The run does not exclude access channels outside that class.

What remains is a **dependency-resolved theory of generative structure**, not a guessed field equation.

2.1 Relation to existing work, and what is new

What is new. This section is a positioning map, not a literature review: no systematic literature search has been performed for this record. Most of the mathematics used in Layers I–IV is standard. The program’s own verdicts say so, grading the machinery “standard mathematics” or NULL-REDUNDANT as a new principle. What the program adds is in-house, pre-registered

verification within declared classes, and the dependency map: which structures the tested models generate, which they do not, and where each supplied input enters. Program-specific in-class findings have not been checked against the literature for priority.

Record section	Standard ingredient it re-demonstrates or relates to	What the program says it added
§4 elimination and memory	Nakajima–Zwanzig and Mori projection [Nak58, Zwa60, Mor65]; Caldeira–Leggett baths [CL83]; Bernstein–Widder completely monotone class [†Ber29, †Wid41]; Kronecker/Hankel minimal realization (named in the instrument); infinite harmonic lattices [†FKM65]	an in-class finite-local demonstration of the chain from locality to memory, continuum, and effective irreversibility
§4.3 irreversibility	Feynman–Vernon factorization [FV63], Poincaré recurrence, Pusz–Woronowicz passivity [PW78], Nakajima–Zwanzig deletion	the organization of these ingredients, per the record’s own statement
§6 influence cone	Caldeira–Leggett-class Gaussian realizability; Feynman–Vernon influence functional [FV63, CL83]; fluctuation–dissipation [CW51, Kub66]	“borrowed-standard” per the P-2 charter; the program adds verification, minimality probes, and the mapping onto the record
§6.4 hierarchy positivity	positivity of the state on the operator algebra generated by the coupling (GNS/Wightman-type positivity [†SW64])	NULL-AS-NEW-PRINCIPLE; one Gram condition organizes the cone’s two inequalities as low-order faces
§8 access	bicommutant and central-decomposition structure; Lanczos chains [†Lan50]	NULL-REDUNDANT as principles; the content is which distinctions each access level makes recoverable
§9 geometry	spectral geometry and “hearing the shape” [†Kac66]; effective resistance [†KR93]; star–mesh (Y– Δ) reduction; Lanczos/recursion [†Lan50, †HHK72]	the eliminations and non-eliminations each access level produces
§10 units	dimensional analysis	the operational discriminator separating a unit change from a physical deformation
§13–§14 probe, universality	Weinberg’s soft-graviton and spin-2 universality arguments [†Wei64, †Wei65]; Gupta-type gauge arguments [†Gup54]	the exact boundary of what is derived given the supplied probe
§16 retained sector	Goldstone modes [†Gol61]	a class selection relative to tested candidates
§17 matter–graviton channel	golden-rule emission; the standard minimal coupling $-\frac{\kappa}{2}h_{ij}T^{ij}$	within-class exponent evidence
§18 TT channel	transverse-traceless projection (textbook transversality)	a classification of channel existence by relative dispersion
nearest relatives	stochastic gravity [HV08]; Sakharov induced gravity [Sak67]; Jacobson’s thermodynamic derivation [Jac95]	comparisons, not support
obstructions	Weinberg–Witten [WW80]	carried as a named-exit condition (Section 22)

References marked † are supplied for this edition. Their bibliographic data have not been verified against the program’s source register (see References).

3. The formal architecture

3.1 The seven principal objects

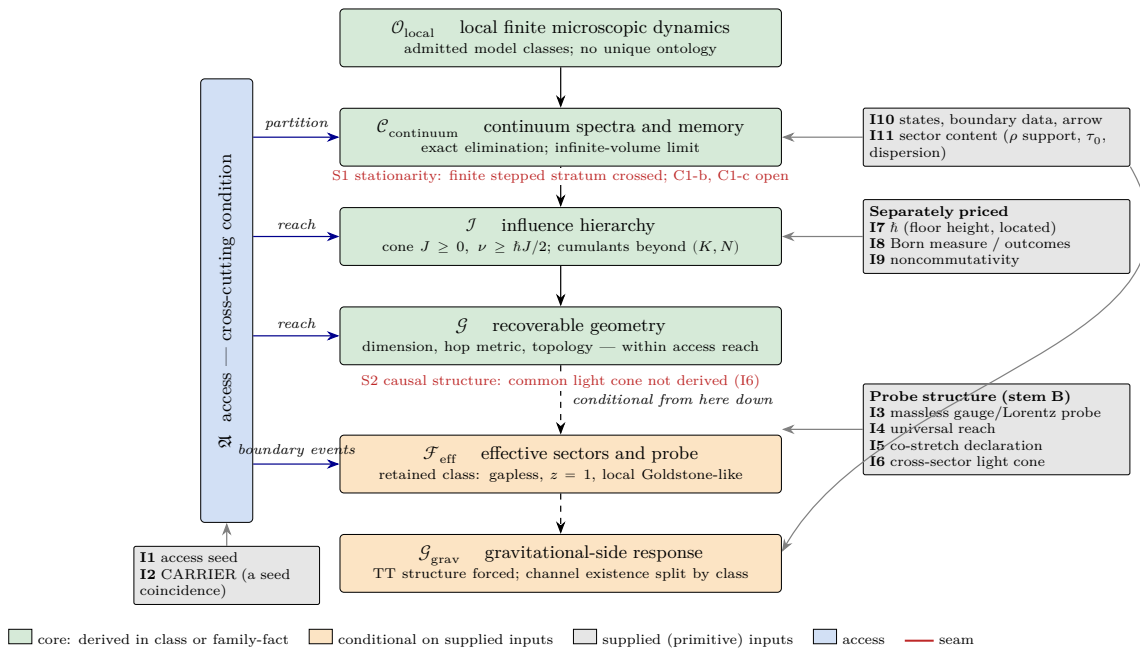
1. $\mathcal{O}_{\text{local}}$: **local microscopic dynamics.** Finite local dynamical systems in the admitted classes (Section 3.3). No unique substrate ontology is selected.
2. $\mathcal{C}_{\text{continuum}}$: **continuum structure.** Spectral and memory structure obtained by eliminating hidden modes and taking limits.
3. $\mathcal{I}$: **the influence hierarchy.** Response, noise, and higher connected correlations of the retained sector's environment.
4. $\mathcal{A}$: **access.**
 - The *seed* is the set of system operators through which an observer or probe couples to the environment (for example, the coupling operator B together with the identity).
 - Its *closure* is the operator algebra the seed generates under products and time evolution.
 - *Boundary events* are changes to the seed.
5. $\mathcal{G}$: **geometry.** Geometry reconstructed from influence and access data by spectral, resistance, metric-density, and walk-count functionals: *geometry as far as access reaches*.
6. $\mathcal{F}_{\text{eff}}$: **effective sectors and probes.** Sector classes inside the influence cone, candidate retained sectors, and the probe structures that couple to them.
7. $\mathcal{G}_{\text{grav}}$: **gravitational-side response.** The conditional gravitational influence point, the transverse-traceless (TT) response, channel conditions, and the exponent class.

3.2 The cross-cutting role of access

Access is consumed in three roles:

- **partition:** it determines which subsystem is retained;
- **reach:** it determines which distinctions can be reconstructed;
- **boundary events:** changing access turns hidden differences into observable ones.

In the tested class, the closure is canonical given a seed, but the seed itself is not dynamically selected (P-5, P-6).



3.3 The admitted model classes

Every result in this record holds within a declared class. The classes actually used are:

Class	Dynamics	Used in
(a) First-order relaxational networks	$\dot{x} = -\mathbf{K}x$, with $\mathbf{K}$ symmetric positive (semi)definite and local (overdamped springs plus on-site pins; RC-type)	persistence checks, minimal-ontology C1–C3, C1-a, C1-a2
(b) Conservative harmonic lattices	$\ddot{x} = -\mathbf{K}x$ on nearest-neighbour lattices, or an oscillator coupled to M harmonic modes	continuum origin, infinite-bath limit, irreversibility I1–I2
(c) Gaussian (quadratic bosonic) baths	star baths with per-mode Gaussian states, classical or quantum	P-2, D-1, G-1, G-2, GS-1, SX-1, CA-1, RS-1
(d) Finite spin systems	tensor products of qubits with local Hamiltonians	P-3, P-4, P-5, P-6, S4-1, and the quantum legs of G-2
(e) Declared continuum sectors	dispersive fields $\omega(p)$ with tree-level (golden-rule) pair channels	CP-1, EQ-1, TT-1; the ω^7 coupling instrument (a quantum harmonic chain coupled to TT gravitons)

Class matters for form. Classes (a) and (b) give memory kernels of different form, and the record keeps them apart:

- In class (a), memory kernels are *completely monotone*: sums of decaying exponentials.
- In class (b), kernels *oscillate*: sums of cosines, whose envelope decays only in the continuum limit.

The admitted principles for “derivable” are: strict locality; finite microscopic content per site; first- or second-order autonomy; passivity; time-translation invariance of the microscopic law (relaxed in Section 5); and standard open-system reduction.

4. Layer I — local microscopic dynamics to continuum

Primary sources. The foundations record of the adjudicator-track branch, archived byte-exact in `archive/adjudicator-track_90218f5/`:

- GRUT Skeleton v02.1 (`GRUT_SKELETON_01.md`, commit 90218f5);
- the instruments `calc/u3_*.py` and their result files, first committed at `affdf52` (24 September 2026) and partly repaired at `5f5395e` the same day.

The foundations record predates the charter protocol of the later campaign. Its checks are written into the instruments’ code; no charter was committed separately before these runs (Appendix D). Section 4.5 gives its full scorecard, including its failed checks.

4.1 Exact finite reduction: where memory comes from

Setting (class (a), Section 3.3). Take a finite local network $\dot{x} = -\mathbf{K}x$, with $\mathbf{K}$ symmetric positive definite. Split the coordinates into one retained coordinate q (site 0) and hidden coordinates y . Write:

- K_{SS} for the retained site’s own stiffness;
- v for the vector coupling the hidden sites to q ;
- $\lambda_l > 0$ and e_l for the eigenvalues and eigenvectors of the hidden block $\mathbf{K}_{yy}$;

- $u_l = e_l \cdot v$ for the coupling of q to hidden eigenmode l .

Eliminating y exactly gives

$$\dot{q}(t) = -K_{SS} q(t) + \int_0^t k(t-s) q(s) ds + \eta(t), \quad k(\tau) = \sum_l u_l^2 e^{-\lambda_l \tau} = v^\top e^{-\mathbf{K}_{yy} \tau} v.$$

Here $\eta(t) = \sum_l u_l e^{-\lambda_l t} m_l(0)$ carries the hidden modes' initial data ($m_l = e_l \cdot y$); it vanishes when the hidden sites start at rest. This is the form the program's instruments implement (`calc/c1_seam.py`; C1-a charter, §1). The memory kernel k is a positive sum of decaying exponentials, so it is *completely monotone*.

Notation note. Earlier program documents write the same kernel as $K_R(t) = \sum_k (v_{1k})^2 e^{\lambda_k t}$ with $\text{Re } \lambda_k < 0$ (eigenvalues of the generator $-\mathbf{K}_{yy}$), and write the reduced equation schematically as $\dot{q} = -\int K_R q + \text{drive}$. That is the same kernel with the opposite sign convention for λ ; the schematic equation omits the instantaneous term and the sign of the memory term. This record uses the decay-rate form above throughout (Appendix I).

Worked example: one hidden node. Let $\mathbf{K} = \begin{pmatrix} K_{SS} & -g \\ -g & \gamma \end{pmatrix}$, so the hidden node has total stiffness γ and couples to q with strength g . Then $v = g$, the hidden propagator is $e^{-\gamma \tau}$, and

$$\dot{q}(t) = -K_{SS} q(t) + \int_0^t g^2 e^{-\gamma(t-s)} q(s) ds.$$

The retained coordinate now has memory with a single time scale $1/\gamma$. As $\gamma \rightarrow \infty$ the hidden node follows q instantly and the memory disappears. This is what “memory exists if and only if there is persistent auxiliary state” means. The foundations record's check P3 uses exactly this model and records the kernel $g^2 e^{-\gamma t} \Theta(t)$.

What the foundations record establishes, in class.

- *Necessity.* Without persistent auxiliary state there is no memory (checks P1 and P2 of `u3_o_rigin_persistence.py`). These are algebraic statements; P1 is asserted in the code rather than computed.
- *The constructive converse* (checks P3, P4, and E4a):
 - Eliminating hidden modes produces the kernel exactly. The repaired P3 records direct kernel extraction to 1.8×10^{-5} and trajectory equivalence to 1.4×10^{-16} . In that model the auxiliary variable realizes the eliminated node exactly, so the second figure is floating-point agreement between algebraically identical updates. The elimination identity is exact algebra; C1-a re-demonstrates it on a 24-site network to 10^{-12} (Section 5.1).
 - The *realization dimension* (the minimal number of hidden modes that reproduces a kernel, i.e. its Hankel rank, by Kronecker's theorem) equals the number of **distinct** eliminated modes: $\{1:1, 2:2, 3:3, 5:5\}$, and a degenerate-pair control returns $M - 1$. P4 synthesizes its kernels directly as sums of exponentials rather than by eliminating a bath model, so it tests realization, not elimination.
 - A pure delay is the limit of an n -stage Erlang cascade. Convergence holds for the integrated kernel (the cumulative distribution), not pointwise: L^1 errors 0.486, 0.376, 0.277, 0.198 for $n = 2, 4, 8, 16$ (E4a).

Repair history (disclosed). P3, P4, and E4a first **failed**: P3 with tail error 0.995 against a threshold of 0.05; P4 with Hankel rank 4 at $M = 5$; E4a with errors that *grew* ($0.721 \rightarrow 1.193$) while its summary said they decreased. The failures were diagnosed as instrument defects:

- P3 had recorded the closed-loop rate instead of the kernel;
- P4 had a variable-shadowing bug and near-degenerate random time constants;
- E4a compared a density against a step.

The instruments were repaired the same day under the owner’s authorization (repair set `5f5395e`), and the converse was re-banked by measurement. The repair was not separately chartered, and the failing result files were regenerated in place at the branch head; the originals remain in git at `affdf52` and are quoted in `U3_RECORD_NOTE_02`. Appendix C.1 states how this record distinguishes an instrument repair from a failed gate, and Section 23.1 lists the failures.

4.2 Continuum emergence (class (b), conservative lattices)

The continuum-origin computation (`calc/u3_continuum_origin.py`) uses class (b): nearest-neighbour periodic lattices with unit springs and unit masses and **no damping anywhere**. The local memory kernel at the origin site is

$$K_0(t) = \sum_k w_k \cos(\omega_k t), \quad w_k = |v_{0k}|^2, \quad \omega_k^2 = \text{eigenvalues of the lattice Laplacian},$$

and in the continuum limit $K_0(t) = \int \cos(\omega t) d\mu(\omega)$, with μ the local density of states. These kernels oscillate. They are not of the completely monotone class (a) form, whose continuum limit is $\int e^{-t/\tau} d\rho(\tau)$ with $d\rho \geq 0$ (Section 4.4).

The computation recorded six checks, all marked as passed, with the verdict `continuum_emergent_from_local_finite_ontology`:

Check	Content	Recorded measurement
(i)	Finite chains converge to the continuum kernel on a fixed time window	Sup gap 0.515 at $L = 16$; 2.8×10^{-15} at $L = 64$; 3.1×10^{-15} at $L = 1024$
(ii)	Mode spacing scales as inverse volume at fixed lattice spacing	$\delta\omega \cdot L = 3.98, 4.33, 4.42, 4.44$ at $L = 16, 64, 256, 1024$
(iii)	Dimension sets the late-time envelope	Envelope slopes -0.52 ($d = 1$), -1.04 ($d = 2$), -1.36 ($d = 3$), against $t^{-d/2}$ targets $-0.5, -1.0, -1.5$. All three kernels change sign in the tail (25, 13, 11 sign changes)
(iv)	Every substrate is nearest-neighbour local	Maximum coupling range 1 (true by construction; asserted in the code)
(v)	The low-frequency class is universal across microscopic spectra	Density-of-states exponents 2.15 (uniform 32^3 lattice) and 1.79 (weakly disordered lattice); Debye value $d - 1 = 2$
(vi)	Viability of gravity as the bath	Order-of-magnitude count only (asserted in the code): suppressing recurrences beyond the cosmic age needs $\gtrsim 10^{18}$ modes; the gravitational field has $\sim 10^{183}$ in a Hubble volume

How to read the table.

- *Five computational checks and one count.* Check (vi) is recorded as a pass but is an estimate, not a derivation; gravity-as-bath remains a hypothesis (Section 12). Check (iv) is true by construction.
- *Check (i) reflects finite propagation speed.* On a fixed window, a long enough chain reproduces the infinite-chain kernel exactly (already at $L = 64$) before boundary reflections return. Finite systems still recur, on a time scale that grows with L .

- *Check (iii).* The instrument labels $d = 1$ “recurrent, oscillatory (J_0 class)” and $d \geq 2$ “power law”. The recorded slopes show the same $t^{-d/2}$ envelope law in all three dimensions, with dimension setting the exponent. The $d = 1$ fit window is recorded as *not* above the finite-size floor, and the $d = 3$ slope deviates from theory by 0.14.
- *Check (v).* The result data record the disordered lattice as 14^3 ; the instrument’s summary line says 12^3 .

Scope. Continuous spectra, $t^{-d/2}$ envelopes, and effective irreversibility in infinite harmonic lattices are classical results [†FKM65]. The computation re-demonstrates them from strictly local finite models, with no damping inserted. It selects no universal spectral content: the density μ (or ρ in class (a)) is supplied. The formal grade of the infinite-volume results is **[FAMILY-FACT]**: demonstrated across the tested families, not a limit theorem.

(Label note: checks (i)–(vi) are internal to this computation. They are unrelated to the completion problems C1–C7 of Section 24.)

4.3 Irreversibility

Two foundations instruments bear on irreversibility.

- **Infinite bath (4/4).** An oscillator coupled to M Hamiltonian modes, with no damping inserted anywhere, acquires dissipative response as $M \rightarrow \infty$ (tested to $M = 2048$). Verdict: `dissipation_emergent_in_thermodynamic_limit`.
- **Irreversibility origin (3/4).** This battery asks whether coarse-graining plus a boundary condition alone can generate irreversibility. Its answer is **no**:
 - a finite closed Hamiltonian system, even with a low-entropy initial condition, only dephases, and its kernel recurs;
 - the verdict is `B_irreducible_dissipation_or_infinite_bath`: effective irreversibility requires either the infinite-volume limit or irreducible microscopic dissipation. GRUT’s claim uses the former.

The failed check tested whether the sign of the absorptive spectrum $\chi''(\omega)$ distinguishes dephasing from dissipation. For the resistive control it recorded a positive fraction of 0.0, while its summary line asserts strict positivity. Preparing this edition found three further problems that the foundations record does not itself disclose:

- the instrument’s sign convention for χ'' may produce the recorded 0.0 (a code observation, unconfirmed because the instrument was not re-run);
- its first two cases run the same simulation;
- the second case’s pass condition is partly vacuous.

Status.

- The **existence** of effective irreversibility is **[FAMILY-FACT]**, as a coarse-grained response in the infinite-volume limit. It is not a proof that microscopic dynamics stop being reversible.
- Its **direction**, the arrow of time, is **[SUPPLIED]** boundary data (inventory item I10).

4.4 Spectral form and uniqueness limits

In class (a) the generator is symmetric and positive definite. All poles are then real and simple, and the stationary kernel has the completely monotone representation

$$k(\tau) = \int e^{-\tau/\theta} d\rho(\theta), \quad d\rho \geq 0$$

(the Bernstein–Widder class [†Ber29, †Wid41]). Passivity in the broader positive-real sense does not by itself exclude complex poles: the class (b) kernels of Section 4.2 oscillate.

Uniqueness has limits. The measure ρ is uniquely recoverable from k only for light tails (a Stieltjes moment-problem condition, named in the instrument). The heavy-tailed kernels relevant to the gravitational constructions lie outside that uniqueness class, although the support of ρ stays observable.

In summary:

- the **form** of the representation is derived in class;
- the **content** of ρ , including its support and any macroscopic scale such as τ_0 , is supplied (inventory item I11).

4.5 The foundations record as a whole

The Skeleton cites some foundations instruments and not others. The full record on the branch is:

Instrument	Battery	Notes
persistence origin (P1–P7)	7/7	5/7 before the same-day repair (Section 4.1)
spectrum (E-series)	13/13	12/13 before repair (E4a)
realization dimension	8/8	
continuum origin	6/6	two checks true by construction (Section 4.2)
infinite bath	4/4	
irreversibility origin	3/4	failed check and further defects in Section 4.3
minimal generative ontology	5/6	the failed check (a derivative coupling outside the positive class) contradicts its own summary; a second check passes although its measured exponent is -0.19 against an expected -0.5
gravity-bath spectral match	8/9	discrete-box kernel correlation $0.43 / 0.29$ against a threshold of 0.98 ; the committed result still reads t^{-3} where the corrected script says t^{-4}
coarse-graining universality	2/7	not cited by the Skeleton; several summaries contradict the recorded branch data
first coupling instrument	5/9	superseded by the rebuilt instrument (Section 17)
kernel minimality; KMS spectral constraint; scale origin; resistive scale; gravitational clock; geometry spectral selection	5/5; 9/9; 7/7; 4/4; 4/4; 9/9	

During preparation of this edition, the six pure-standard-library instruments were re-executed from a copy: spectrum, realization dimension, geometry spectral selection, kernel minimality, KMS spectral constraint, and scale origin. All six reproduced their committed scores. The instruments that need `numpy` were not re-run.

5. The stationarity seam and its current resolution

Why stationarity matters. Section 4 assumed that the environment’s memory depends only on the lag $t - s$ (stationarity). For a free graviton-like field on the standard expanding-universe background, a calculation in the cosmological record shows that it does not.

The measurement (non-stationarity instrument, 7/7, 5ea380e; governed by owner Correction 01, 6fc5139). The object is the exact retarded (commutator) kernel of a free transverse-traceless (TT)

tensor mode at fixed comoving wavenumber k on the declared Λ CDM background. It was sampled at anchor redshifts $z_a \in \{0, 0.25, 0.5\}$, lags $0.1, 0.2, 0.4/H_0$, and $k = 0.5, 1, 2$. On that frozen grid:

- The best unconstrained Δt -only approximation leaves a pooled residual fraction $R = 0.516$ of the kernel’s sampled variation.
- The same-lag drift reaches 1.53: at fixed lag, the kernel changes by more than its own mean magnitude as the anchor moves. Part of this is a fixed-comoving- k labelling effect, which a de Sitter comparator also shows (drift ≈ 0.99). The excess specific to the expanding background is 0.17, 0.24, and 0.54 at lags $0.1, 0.2$, and $0.4/H_0$.
- **Kernel transport** (11/11, 05226bf). No tested local transport rule (the de Sitter kernel re-rated at the observation, midpoint, or emission rate) reproduces the exact kernel over the relevant interval. The local de Sitter transport matches to $\leq 0.3\%$ at lags up to about $0.25/H_0$; the best member is about 3% off at $0.55/H_0$; all are off by 34–130% by $0.8/H_0$.

Scope. This is a measurement on the sampled domain, not a theorem. It does not establish that a stationary reduction is impossible in another representation (for example, the smeared worldline correlator), for another observable, or in another regime.

The finding created **seam S1**: the continuum map ϵ of the core was [UNDEFINED] off stationary backgrounds.

5.1 C1-a — the finite stepped nonstationary extension

C1-a (charter fa6ac44, verdict 55a708d, battery 10/12) attacked S1 by *generating* a two-time kernel from a declared nonstationary microscopic law. The cosmological kernel was never used as the model.

The model (class (a)). A chain of 24 sites with unit springs and pins 0.3 on every site; site 0 is retained and the other 23 are hidden. Time is split into epochs $[0, 4)$, $[4, 8)$, $[8, 12]$. During the middle epoch the four hidden springs $(1, 2), \dots, (4, 5)$ are multiplied by $1 + \epsilon_m$, with $\epsilon_m \in \{0, 0.01, 0.02, 0.5\}$. The coupling spring $(0, 1)$ is never modulated, so all nonstationarity lives in the hidden network. The retained site starts excited and the hidden sites start at rest, so $\eta = 0$.

Why the kernel becomes two-time. Within one epoch e the hidden block has eigenvalues $\Lambda_e = \{\lambda_{e,l}\}$, eigenvector matrix V_e , and couplings $u_e = V_e^\top v$. Across a step at time t_* (a sudden quench),

$$k(t, s) = u_B^\top e^{-\Lambda_B(t-t_*)} C e^{-\Lambda_A(t_*-s)} u_A, \quad C = V_B^\top V_A,$$

for s before and t after the step. The matrix C is the overlap between the old and new hidden eigenmodes. The kernel is not a function of $t-s$ alone, and it reduces to $\sum_l u_l^2 e^{-\lambda_l(t-s)}$ when $\epsilon_m = 0$ ($C = \mathbb{1}$).

The reduced solve. The instrument solves the retained dynamics using only retained-level (“S-level”) data: the per-epoch spectra $\{\lambda_e, u_e\}$, the mixing matrices C , and K_{SS} . Within an epoch, $\dot{m}_l = -\lambda_l m_l + u_l q$ and $\dot{q} = -K_{SS} q + \sum_l u_l m_l$; at each boundary, $m \mapsto C m$. Integration is fourth-order Runge–Kutta with step 0.002.

Component	Result
The map ϵ	DERIVED EXTENSION, in class. The S-level solve reproduces the exact full nonstationary dynamics to 1.0×10^{-12} . The step-halving error ratio is 17.85, a fourth-order signature, so the residual is integrator error.

Component	Result
Continuity	$k(t, s) \rightarrow k(t - s)$ exactly when time-translation invariance is restored (7.8×10^{-16}). The deviation is linear in the modulation amplitude (ratio 1.9968 between $\epsilon_m = 0.02$ and 0.01).
Earned structure	Preserved. Passivity holds, and the multi-time Gram matrix stays positive semidefinite off the stationary domain (to 8.6×10^{-14}). A deliberately tampered Gram matrix is detected at -0.160 . Hierarchy positivity is therefore stationarity-independent by measurement, not only by identity.
Packaging	Two frozen gates RED (Section 23): the same-lag drift measured 0.0498 against a threshold of 0.1, and the local-anchor family’s midpoint member measured 0.0494 against 0.05. The labeled diagnostic found the thresholds miscalibrated against the frozen <i>peak</i> normalization: $k(0) = v^T v$ does not depend on the modulation, so normalizing by the peak suppresses the effect being measured.

The owner accepted the map, continuity, and structure components and carried the packaging question to C1-a2. The formal domain extension is recorded as Formalization 01 Amendment 02 (equivalently, Formalization 02 Amendment 01).

5.2 C1-a2 — the packaging boundary, certified

C1-a2 (charter 3b139ab, verdict 4deda60, battery 7/7) is a separately chartered re-test. C1-a’s record was declared immutable, and its two reds stay red. “Packaging” means representing $k(t, s)$ by a single stationary spectral measure or by a single function of $t - s$.

Design. C1-a’s model, with two working points ($\epsilon_m = 0.5$ and 1.0) and a stationary control. Anchors: three times in the first epoch (1.5, 2.0, 2.5) and three in the modulated epoch (5.5, 6.0, 6.5). Lag window $\tau \in [0.5, 3.0]$. The statistics use the kernel’s own local scale at each lag instead of its peak.

Calibration disclosure. The charter states that its thresholds were set with roughly a $2\times$ margin below C1-a’s labeled diagnostic values: “calibration by prior measurement, disclosed — not blind prediction”. Its recorded strength is “freshly frozen, mechanically evaluated”.

The gates below are renamed A2-1 to A2-6; the charter calls them P-1 to P-6, labels unrelated to the forks P-1 to P-6 of Sections 6–8.

Gate (charter label)	Statistic	Result	Threshold
A2-1 (P-1)	Largest local-scale difference between the kernel at the same lag from anchors in different epochs, $\epsilon_m = 0.5$	0.1182	> 0.05
A2-2 (P-2)	Ratio of that statistic at $\epsilon_m = 1.0$ to $\epsilon_m = 0.5$	1.6769	1.5–2.5
A2-3 (P-3)	Worst error of each frozen local-anchor rule (anchor at emission, observation, midpoint), $\epsilon_m = 1.0$	0.1095 / 0.1484 / 0.0968	each > 0.05
A2-4 (P-4)	Largest per-lag spread of $k(a + \tau, a)$ over the six anchors, divided by its mean	0.1100	> 0.05
A2-5 (P-5)	Stationary control, both statistics	4.4×10^{-15} , 1.8×10^{-15}	$< 10^{-10}$
A2-6 (P-6)	Stationary world with all springs $\times 1.25$: drift statistic	3.8×10^{-15}	$< 10^{-10}$
A2-6 (P-6)	The same world: kernel difference from the base world	0.2737	> 0.05

A2-4 excludes the whole class. At each lag, the per-anchor mean is the L^2 -optimal Δt -only fit.

A2-4 therefore excludes every Δt -only kernel at the certified scope, not just three prescriptions.

A2-6 fixes the statistic's meaning. A stationary world whose kernel differs by 0.2737 gives zero nonstationarity signal. The statistic detects nonstationarity, not kernel difference.

Certified at finite stepped level, in class: no single spectral measure and no single Δt -kernel packages the generated $k(t, s)$ over the sampled anchors and window. The exclusion is not a theorem over all nonstationary laws. The replacement datum is exhibited constructively:

$$[\{\lambda_e, u_e\}_e, \{C_{e'e}\}_{e,e'}], \quad C_{e'e} = V_{e'}^T V_e,$$

or equivalently the family of hidden-network propagators. **Minimality is not proved.** The owner accepted C1-a2 as Formalization 02 Amendment 02 (6abbf316), the source boundary of this record.

Remaining parts of S1:

- **C1-b:** the infinite-volume nonstationary limit (open);
- **C1-c:** smooth modulation (open).

6. Layer II — influence data

6.1 Definitions, and the Gaussian influence cone (P-2; 16/16)

Definitions (the P-2 class). A system couples linearly, through an operator B , to a Gaussian bath of harmonic modes μ with frequencies $\omega_\mu > 0$ and couplings c_μ . The bath's entire effect on the system (its Feynman–Vernon influence functional [FV63]) is then fixed by two kernels:

$$K(t) = \sum_\mu J_\mu \frac{\sin \omega_\mu t}{\omega_\mu} \quad (\text{dissipation, or response}), \quad N(t) = \sum_\mu J_\mu s_\mu \frac{\cos \omega_\mu t}{\omega_\mu} \quad (\text{noise}).$$

Here $J_\mu = c_\mu^2$ is the spectral weight of mode μ , and s_μ is its symmetric occupation: $s = \frac{1}{2}$ in the vacuum and $s = \frac{1}{2} \coth(\hbar\omega/2k_B T)$ at temperature T . The spectral forms are the **dissipation spectrum** $J(\omega)$ and the **noise spectrum** $\nu(\omega) = \hbar s(\omega)J(\omega)$. The P-2 instrument works in units $\hbar = k_B = 1$; the factor $\hbar$ here is restored by declaration, so that the vacuum sits on the floor below.

Within this class, the realizable pairs form the cone

$$\mathfrak{C}_{\text{Gauss}} = \{(J, \nu) : J(\omega) \geq 0, \quad \nu(\omega) \geq \frac{\hbar}{2} J(\omega)\}.$$

- $J \geq 0$ is the passivity half. It is automatic for real couplings ($J_\mu = c_\mu^2$).
- $\nu \geq \hbar J/2$ is the quantum half. It says $s \geq \frac{1}{2}$: the mode occupation above the zero point is non-negative.

Worked example: a thermal bath. At $T = 0.7$ (units $\hbar = k_B = 1$) the instrument's frequency grid runs from $\omega \approx 0.51$ to 2.09, and $s(\omega) = \frac{1}{2} \coth(\omega/1.4)$ runs from 1.43 down to 0.553. Every value exceeds $\frac{1}{2}$, so the thermal state lies strictly inside the cone. As $T \rightarrow 0$, $s \rightarrow \frac{1}{2}$ and the state reaches the floor: the vacuum saturates it exactly (reconstruction to 2.3×10^{-16} in K and 1.5×10^{-16} in N).

Minimality was probed constructively:

- the vacuum saturates the floor exactly;
- the interior is filled, including a genuinely non-thermal (non-KMS) occupation profile;

- three proposed additional constraints were each defeated by an explicit realizer.

KMS, the fluctuation–dissipation theorem (FDT), stationarity, and single-pole memory are **states inside the cone, not constraints on it**. None of them selects the memory spectrum.

Class boundary. The cone holds for the declared class: harmonic modes of positive frequency in Gaussian states, one scalar channel. Outside it, for example in gain media or with effective negative-frequency oscillators, $J < 0$ occurs; the general quantum bound is the standard $\nu \geq \frac{\hbar}{2}|J|$.

Provenance. The P-2 charter declares the cone “borrowed-standard Gaussian open-system mathematics (Caldeira–Leggett-class realizability)” [CL83, CW51, Kub66]. P-2’s contribution is instrument-grade verification, the minimality probes, and the mapping onto this record. It is not a GRUT derivation.

6.2 What the cone explains, and what it does not

The floor $\nu \geq \frac{\hbar}{2}J$ locates $\hbar$ geometrically, as the height of the fluctuation floor. It does not derive the value:

$\hbar$ is located, not generated.

Classical physics is the same cone with the floor removed. This does not explain why the physical world is on the quantum branch.

6.3 Why (K, N) is insufficient (P-3; 24/25)

P-3 built two genuinely different spin baths with exactly matched two-point data: both have two-point function $C(t) = 2g^2 e^{-i\omega_0 t}$, where g is the bath’s coupling strength, and both have vanishing third cumulants. Their connected fourth cumulants differ: $\kappa_4 = -4g^4$ against $-8g^4$. A probe qubit’s coherence (maximum 1) evolves differently in the two baths, by up to **0.338**; a matched control reads 0.0.

So (K, N) is a projection of the influence structure, not all of it. The first physical obstruction appears at the first unmatched cumulant, and P-4 constructs pairs that first differ at orders 4, 6, and 8. **Matching to finite order never certifies equivalence.**

The matrix lift. For several coupling operators B_a , define the matrix spectrum $C_{ab}(\omega)$ of the bath correlations, and $\nu \equiv \frac{1}{2}(C(\omega) + \bar{C}^\top(-\omega))$, $J \equiv C(\omega) - \bar{C}^\top(-\omega)$. The candidate lift of the cone, $\nu \pm J/2 \succeq 0$, is then literally $C(\omega) \succeq 0$ and $\bar{C}^\top(-\omega) \succeq 0$: positivity of the bath state (units $\hbar = 1$). P-3 graded it **NULL-AS-NEW-PRINCIPLE**. For one commuting channel it reduces to $\nu \geq |J|/2$.

6.4 The full influence hierarchy (P-4; 21/22)

The **influence hierarchy** is the set of all multi-time correlators $\langle B(t_1) \cdots B(t_n) \rangle$ of the bath coupling operators.

- *Complete as an interface, in class.* In the tested class, two environments with the same hierarchy produce identical dynamics for every observable accessible through B ; P-4 constructed no counterexample. Bounded-moment determinacy is cited as standard mathematics and claimed in class only.

- *Its positivity is state positivity.* The hierarchy is admissible exactly when every Gram matrix of monomials in $\{B(t)\}$ is positive semidefinite, which is positivity of the bath state on the algebra those operators generate:

influence-hierarchy positivity $\equiv$ state positivity

[IDENTITY; NULL-AS-NEW-PRINCIPLE]

One Gram condition contains the cone’s two inequalities as its low-order faces. This is a structural reformulation of quantum admissibility, not a derivation of quantum mechanics. One frozen gate failed (the order-8 window; Section 23).

7. Sector counting and influence exponents (S-1; 18/18)

The exponents. Write $J(\omega) \propto \omega^s$ at low frequency. The record carries three exponent classes, the “3/5/7 ladder”:

- $s = 3$: the matter (acoustic) channel of the earlier record;
- $s = 5$: the flat-space graviton-loop kernel;
- $s = 7$: the two-phonon graviton channel (Section 17).

These values come from different records in different conventions, so they are compared only through their increments.

Counting rules measured in S-1:

- each extra power of momentum in the vertex adds +2.000;
- phase space adds $d - 1$;
- an order-2 symmetry cancellation (tracelessness) adds +3.999.

The ladder is **consistent by mechanism** at the increment level, where conventions cancel. No single formula derives its absolute values.

The counting law, as updated by the P-3/P-4 ruling (provisional). In the tested families, locality and symmetry select **branch** exponent classes. At matrix level the observable exponent then follows spectral competition among branches:

locality + symmetry $\rightarrow$ branch classes $\rightarrow$ spectral/matrix selection $\rightarrow$ effective sector.

Two further results bound the law:

- **No tested extremum selects the exponent.** None of the three declared extremization functionals (memory duration, low-frequency fraction, zero-point load) selects any recorded exponent. The one band-robust interior extremum found ($s = 1.25$) lands on nothing physical. Amplitudes stay free.
- **Matrix masking.** With several channels, the lowest exponent dominates at low frequency (“min-dominance”), so a cancelled branch can be hidden by another. The frozen eigenvalue-level gate for S-1’s hypothesis H3 remains **RED** (+2.216 against a predicted +4). The labeled post-hoc diagnostic shows the +4 increment surviving at branch level: +3.999 exactly on the cancelled branch’s spectral weight.

Locality and symmetry constrain the scaling class of a sector’s influence data. They do not select its amplitude, state, or microscopic realization.

The spectral-density classification $J \propto \omega^s$ itself is standard [Leg87]; S-1 grades its own law “a compatibility principle, not a new law”.

8. Layer III — access as a physical interface

The campaign adopts, and its results repeatedly illustrate, the following working principle. It is a definition, not a derived result, in the spirit of operational approaches to the foundations of quantum theory:

A distinction is physically meaningful to an observer exactly to the extent that it changes the influence data accessible to that observer.

8.1 Subsystem selection (P-1; 6/6)

In the tested class (the exactly solvable Gaussian toy class, one 10-site testbed per regime), P-1 found three regimes:

- **Structured worlds:** the dynamics can distinguish a subsystem without help. One intrinsic criterion, causal cohesion, selected the impurity with gap 0.678, and the selection was stable under coarse-graining. No second criterion agreed, so the selection is partial and criterion-relative; the pre-registered signature was not reached.
- **Symmetric worlds:** descriptions are representationally equivalent; the intrinsic criteria return only tie-orbits.
- **Generic worlds:** no partition is selected by the tested criteria.

The natural criteria (bath rank, minimal entanglement, causal cohesion) disagree with one another; no universal subsystem selector was found. The instrument’s mechanical signature printed “B”, and the pre-registered three-way taxonomy proved incomplete. The trichotomy is the verdict’s reading of the per-testbed data, and the owner ruling accepted it on that basis.

8.2 Seed versus closure (P-5, 22/22; P-6, 38/39)

The **seed** is the set of operators through which a probe couples; its **closure** is everything generated from the seed and the Hamiltonian under products and commutators (Section 3.1).

- **Given a seed, its closure is canonical.** The seed itself matters: the same dynamics with seed $\{\mathbb{1}, B\}$ alone closes at dimension 2, not 4.
- **The map from seeds to closures is not injective (P-6).** On a 3-spin chain, the two end seeds σ_z^1 and σ_z^3 share the causal graph and the conservation law, and each generates exactly the same 32-dimensional parity-commuting algebra. Yet a probe coupled through one or the other sees coherence curves differing by **0.287**.
- **Minimal sufficient seeds depend on the elimination path.**

That seeds are organized by the algebra’s central decomposition is standard structure (NULL-REDUNDANT as a principle). **The access seed is a supplied input (I1).** One frozen P-6 gate failed (Section 23).

8.3 Boundary events (P-5)

A probe qubit couples to spin 1 of a three-spin network through $B = 0.5 \sigma_x^1$ (pure dephasing). Two hidden environments, A and B, agree on every multi-time correlator of B :

- the probe coherence is identical to 1.3×10^{-14} over $0 < t \leq 10$;
- yet the environments differ outside access by 1.547 in $\langle \sigma_z^2(t) \rangle$ (maximum possible 2).

At $t_* = 3.0$ the coupling is extended to $B + 0.4 \sigma_x^2$, a change of access. Before the change the probe coherences agree to 7.8×10^{-16} ; after it they split by **0.38**. The difference is invisible forever under fixed access and consequential exactly when access changes. Access therefore has a concrete physical role, not only a bookkeeping one.

8.4 Medium versus relational descriptions (D-1; 13/13)

Take two non-isomorphic environments: a 9-dimensional star bath, and an 11-dimensional chain bath with two hidden decoupled modes. Their influence data are identical. Every observable of the system alone agrees to 4.5×10^{-15} , and observables routed through a probe coupled only via the system agree to 6.3×10^{-15} .

Conversely, two theories with the same kernel K but different noise N split observably: the same bath in a thermal rather than a ground state changes the noise kernel by 1.09, and the system's trajectories split by 0.91.

Scope. The tested class is exactly solvable Gaussian baths, factorized initial state, one system degree of freedom, finite baths, and access levels as declared. There, at fixed accessible influence data, the question “medium or relational?” is representational. The run does not exclude an access channel outside that class.

Relation to standard results. For linear coupling to a Gaussian bath, that reduced dynamics depend on the bath only through (K, N) is the Feynman–Vernon result [FV63], and the star and chain forms are related by Lanczos tridiagonalization [†Lan50]. D-1's contribution is operational: it fixes the criterion “same influence data implies same physics at this access” on which the rest of the record relies.

9. Layer IV — geometry as reconstructed structure

9.1 G-1 — the arrival-time route (26/34; eight gates RED)

In the tested class and at the tested scales, arrival-time functionals of interface data are not metric on finite dispersive substrates. The located obstructions are dispersive precursor tails, boundary reflections, and impedance mismatch; G-1's held-out distance checks erred by 37%. All eight failed gates remain red (on gated checks alone the battery is 9/17). G-2 later read the failure as one of the arrival-time instrument, not of geometric information.

9.2 G-2 — spectral geometry (35/35)

Spectral functionals of the influence data recovered:

- **dimension** 1.000, 2.000, 2.942, and 1.000 for 1D, 2D, and 3D lattices and a quantum XX chain (true values 1, 2, 3, 1). The frozen estimator is $\hat{d} = 1 + \log_2[S(6)/S(3)]$, where $S(r)$

counts sites at hop distance r ; on the cubic lattice its exact value is $1 + \log_2(146/38) = 2.942$, so that figure is the estimator's, not a numerical error;

- **hop distances** within 0.004;
- exact additive integer distance triples (for example $36 = 15 + 21$);
- **metric density** tracked to 10^{-9} , with a held-out check at 2×10^{-12} .

All reconstructions survive representation scrambles. The recorded grade is GEOMETRY-PARTIAL-SPECTRAL, with a structural GEOMETRY-UNDERDETERMINED component. The machinery (heat kernels, effective resistance [†KR93], Lanczos recursion, multidimensional scaling) is standard. The content is which invariants a declared access makes recoverable. This establishes an in-class reconstruction route, not access-independent geometry.

9.3 Access-relative geometry

Single-site degeneracy. Under single-site access, a 2D grid and a 1D Lanczos chain have identical resolvent data (to 2.2×10^{-16}) and heat-kernel data (to 4.4×10^{-16}). This is a structural degeneracy of the interface, a consequence of the recursion method: any single-site spectral measure is reproduced exactly by a semi-infinite chain [†Lan50, †HHK72].

Topology horizon. Rings of circumference 40 and 80 have closed-walk counts that agree as exact integers through order 38 and first differ at order 40. The first distinguishing order equals the circumference.

9.4 GS-1 — selection by access richness (19/19)

Access	Geometric selection
Full, site-resolved	Selected in class, up to relabeling. Reconstruction at full access is immediate by inversion. The finding is that even an isospectral candidate (a rotated stiffness matrix whose trace moments match to 2×10^{-16}) is eliminated: local data differ by 1.59.
Boundary, dynamic	Dynamic data eliminate the Δ network. Static data are blind: Y and Δ are static-identical to 2×10^{-16} . The first separating term is the ω^2 coefficient, 0.1975, matching the analytic value. The dynamic data select the interior dimension (Y over Δ); the interior geometry itself remains CONSTRAINED , not uniquely selected.
Boundary, static only	Blind
Single site	Underdetermined: a non-isometric family survives every earned test

Terms. Y is a three-terminal star with one interior node; Δ is its star-mesh (Y - Δ) equivalent, with edge weights $w_i w_j / \sum w$. They have identical static response. The first dynamic difference is the ω^2 coefficient $K_{AI} K_{II}^{-2} K_{IA}$, the inertia of the hidden node; the figure 0.1975 is its largest matrix entry (16/81).

Scope. Small quadratic (Gaussian) networks, where the two-point data are the full hierarchy (P-4); the full-access test object is a 3×3 grid. Non-Gaussian sectors could carry more.

The resulting hierarchy is

$$\boxed{\text{influence data} \rightarrow \text{access} \rightarrow \text{spectral structure} \rightarrow \text{geometry}}$$

and not “geometry $\rightarrow$ everything”. Absolute geometry is not currently an object of GRUT.

A correction on the record. A candidate passivity selector, LocPos (every edge weight and on-site pin non-negative), had appeared to select a unique geometry at single-site access. A labeled post-hoc diagnostic showed that with interior pins of 0.1 it admits 96 members, 94 of them non-isometric. LocPos narrows the family; it does not select a member. The original line stays on the record beside the correction.

10. Spatial scaling and units (SX-1; 23/23)

The physical picture. A lattice field with nearest-neighbour springs k and on-site pins p (mass-like terms) has two intrinsic lengths: the lattice step and the correlation length $\ell_c = \sqrt{k/p}$, the lattice analogue of a Compton wavelength. A **pure unit change** is a transformation that leaves every dimensionless accessible datum invariant. $|\Delta\mathcal{D}|$ is the largest change in a declared set of dimensionless accessible data.

- **Co-stretch.** All couplings, pins included, rescale together. Dimensionless data stay invariant to 2.4×10^{-15} . This is a unit change [IDENTITY].
- **Rigid stretch.** Springs rescale while the pins stay fixed, so ℓ_c changes in lattice units. This is observable ($|\Delta\mathcal{D}| = 0.013$). Anisotropic and inhomogeneous deformations are also observable (0.094 and 0.180).

All of these deformations are admitted by the earned structure: it *detects* which one a probe performed but *forbids* none. The statement “spatial rescaling is always a change of units” is therefore false in the tested class, for sectors with two or more intrinsic scales; for single-scale sectors the distinction is vacuous. Which operation a gravitational probe performs is the supplied **co-stretch declaration** (Sel-4x; inventory item I5), the ruler counterpart of “a uniform static field rescales all rods equally”.

Decidability splits by access. A pure unit change and a unit change plus an interior deformation give identical single-site data (to 1.3×10^{-14}) but separate under full access (6.05). SX-1 grades its classification leg “dimensional analysis made operational”.

11. Quantum structure: what is and is not recovered

Recovered, in class:

- the *form* of the quantum fluctuation floor (its height, $\hbar$, is supplied);
- matrix positivity constraints, stronger than channelwise positivity;
- discrimination by higher cumulants;
- the admissible influence hierarchy;
- the classical limit, as the same cone with the floor removed.

Not derived:

- $\hbar$. Located, not derived. The earlier record’s $\hbar$ -emergence attempt failed; its primary computation is not committed in this record’s sources.
- **The Born rule.** The routes tested in the program’s earlier record returned negative results. Decoherence selects a pointer basis; it does not select an outcome. This is not an evaluation of Gleason-type, enviance, or decision-theoretic derivations in the literature.

- **Noncommutativity.** Supplied. The routes tested from a commutative substrate produced no quantum structure. Published programs of that kind were not evaluated.

12. The gravity program

The gravity branch is conditional and downstream of the core. Success in it would constrain a gravitational identification; failure in it would not invalidate Sections 4–11.

The gravitational identification is the program’s central physical hypothesis: that the persistent, memory-bearing continuum of Layer I is physically realized by gravity, so that gravity is the universal dissipative environment to which all matter couples. It is graded **HYPOTHESIS**, not derived. The foundations record found a gravity-like spectral bath viable at 8 of 9 spectral checks (Section 4.5), which is consistency, not derivation. Completion problem C7 (Section 24) is the test that could kill it.

$$(\mathcal{J}, \mathfrak{A}, \mathcal{G}) \rightarrow \mathcal{F}_{\text{eff}} \rightarrow \mathcal{G}_{\text{grav}}$$

The probe. Throughout, the gravitational probe is the metric perturbation h_{ij} . It couples to a matter sector through its stress tensor, $L_{\text{int}} = -\frac{\kappa}{2}h_{ij}T^{ij}$, the minimal universal coupling, with $\kappa \sim 1/M_{\text{Pl}}$. The forks model it in three ways:

- as a uniform strain (CP-1, EQ-1, SX-1);
- as a constant clock field (S4-1, FS-1, U-1);
- as a massless spin-2 field (CC-1, TT-1, and the ω^7 instrument).

In plain terms. Suppose one supplies a massless spin-2 probe that couples universally to a conserved stress tensor, with a common light cone and a declared unit convention (inputs I3–I6, Section 21). Then, within the tested lattice models, the matter-side sector that can absorb or emit it must be gapless, linearly dispersing, and Goldstone-like, and in the tested coupling class its emission spectrum scales as ω^7 in the program’s convention. Nothing in the branch derives the probe or its coupling. This is what the record means by “general relativity is recovered with imports, not derived” (Section 29).

The reduction chain. Each assumption in the chain was attacked in sequence. Each attack reduced the assumption to a named, more specific supplied premise, derived part of it within a stated class, or split it by class:

Fork	Assumption attacked	In standard terms	Outcome
CP-1 (39/39)	Minimal-stress coupling	Which local stress coupling the probe uses	The couplings that give the exponent’s +4 increment form a plane, so the exponent does not identify the coupling. Conservation leaves an improvement freedom ξ ; adding infrared Weyl symmetry selects the canonical tensor in $D = 2$ and the improved tensor in $D = 4$. Reduced to geometric (proper-distance) coupling at $\xi = 0$.
EQ-1 (45/46)	Geometric-only coupling	The weak equivalence principle	Earned selectors leave geometric and non-geometric couplings side by side. Reduced to Sel-4: a constant probe acts on each sector as a pure change of units, invisible in the sector’s own dimensionless data.

Fork	Assumption attacked	In standard terms	Outcome
S4-1 (28/28)	Sel-4 as a whole	Gravitational redshift and its universality	Splits by part and by class. The temporal part (Sel-4t: the probe couples to the sector’s energy, $O \propto H$) is derived in class for generic sectors, given C_cons. The universality part (Sel-4U) is derived for interacting sectors and irreducible for non-interacting ones.
FS-1 (31/31)	The free-sector exception	—	The exception is structural (Section 14).
CC-1 (29/29)	C_cons	Current conservation	Irreducible. Reduced to a supplied massless gauge probe.
U-1 (23/23)	Clock universality	Universality of coupling across sectors	Class split. Derived in class under exchange, conditional on the supplied massless gauge probe; irreducible for decoupled sectors.
RS-1 (27/28)	Retained-sector choice	Which matter sector the probe couples to	A class is selected, conditional on CARRIER and the probe.
CA-1 (23/23)	CARRIER	—	Irreducible. Reduced to an access-seed coincidence.
SX-1 (23/23)	Sel-4x	Uniform rescaling of rods	Irreducible. Reduced to the co-stretch declaration.
TT-1 (9/10)	The physical TT channel	Graviton emission by pairs of matter quanta	ξ has no effect on the physical channel; the tensor structure is forced; existence splits by class.

13. Probe structure and universality

13.1 Conserved-current coupling (CC-1)

C_cons is the requirement that, at zero frequency and zero momentum, the probe couples to a conserved local charge: its operator O commutes with the Hamiltonian, $[O, H] = 0$.

A non-conserved coupling, the phonon mass modulation $O_M = \frac{1}{2} \sum_i p_i^2$, passes every earned selector at the constant-probe level:

- the influence cone;
- hop and resistance geometry;
- static regularity.

It is spectrally a unit change. The criterion “stationary under driving” turns out to be C_cons restated: requiring that a driven probe produce no spectral weight at nonzero frequency is exactly $[O, H] = 0$, an identity, so it adds no independent support. Imposing C_cons wholesale would also delete GR-1’s own T_{xx} pair channel, which does not commute with H ; C_cons can apply only to the probe’s constant (clock) component.

The positivity route separates the cases. Exchange positivity was tested in $D = 4$ with 200 random sources per case (normalization arbitrary; only the sign carries content):

- **Massless probes:** vector and spin-2 exchange become negative for non-conserved sources (minimum residues -10.6 and -17.8).
- **Massive probes:** exchange stays positive.

In the tested cases, C_{cons} follows from the cone for a supplied massless gauge probe, and not for massive probes.

This parallels Weinberg’s low-energy derivation of charge conservation and universal graviton coupling from massless spin-1 and spin-2 exchange [†Wei64, †Wei65]; CC-1 grades itself “the Weinberg/Gupta–Bleuler-type argument”, NULL-REDUNDANT as a new principle. Its contribution is the dependency map. (*The non-uniqueness of the exponent locus, often quoted beside this result, is CP-1’s finding; see Section 12.*)

13.2 Universal reach (U-1)

Given the supplied massless gauge/Lorentz probe structure, exchange-coupled sectors are forced to universal clocks (derived in class). The soft-emission gauge variation in $D = 4$, over 100 random configurations per class, is at least 0.13 for unequal clock couplings and at most 9×10^{-16} for equal ones. The forcing comes from the supplied structure, not from anything earned; U-1 grades it “Weinberg’s soft-emission argument: standard”.

For genuinely decoupled sectors that variation vanishes exactly. Non-universal clocks there are observable by joint access (cross-sector discriminator 0.049, against 1.8×10^{-14} for universal coupling), yet nothing earned forbids them.

A dynamical probe that couples only through clocks mediates no exchange: the bath energy $\langle H_B \rangle(t)$ stays constant to 7.5×10^{-15} . Universal reach is therefore not self-generating: **it is supplied** (I4).

14. Clock coupling and the free-sector loophole (S4-1, FS-1)

Let O be the operator that the probe’s constant (clock) component couples to. In general relativity a uniform potential couples to the total energy, $O = H$, which only rescales the sector’s clock rate (gravitational redshift).

- **Generic interacting sectors.** Such a sector has no local conserved charge other than H , so C_{cons} forces $O \propto H$ (derived in class, S4-1; the finite-system analogue of Weinberg’s universality argument).
- **Free and integrable sectors.** An exact phonon ring carries exactly $2R$ independent local conserved quadratic charges within range R , so H is not singled out.

This opens **loophole L1**:

The gravity-side retained sector lies in the exception class of the clock-coupling ($O \propto H$) argument used to motivate its own coupling chain.

The candidate **GeoInv** (“a constant probe must not rewire the sector’s recovered hop geometry”) would, together with C_{cons} , select $O = H$ in the free sector. It is an **unpromoted candidate**, not an axiom. GR-1’s pair channel is *insensitive* to this ambiguity, because conserved clock couplings contribute nothing to it; FS-1 records that insensitivity is not the same as resolution.

15. Carrier selection (CA-1)

CARRIER is a bridging premise introduced in RS-1: *the matter sector the gravitational probe couples to is the same sector whose static response carries the geometry recovered in Section 9.* “Carries the geometry” means that its effective resistance $R(0, r)$ is additive in the separation r .

CA-1 tested whether CARRIER can be derived:

- **The recovered geometry cannot pick the carrier.** Four genuinely different sectors built on the same substrate operator (a phonon, a $z = 2$ Schrödinger boson, a rescaled twin, and a classical relaxational field) recover the same normalized resistance geometry, to 4×10^{-14} . CA-1 notes that this leg is close to an identity: sectors built on the same stiffness operator share its geometry by construction.
- **The cone makes one earned cut.** It eliminates the classical carrier, because at $T = 0$ its noise vanishes and $\nu - J/2 = -0.192$ (units $\hbar = 1$). Three quantum carriers survive. CA-1 records that this cut has an escape: a quantum-completed dissipative field would sit on the floor and survive.
- **A non-carrier bath survives everything earned.** Free fermions pass every earned test.

CARRIER is therefore not derived. It reduces to an access-seed coincidence: *the sector the gravitational probe couples to is the sector the geometry was recovered from*, i.e. the probe’s access seed coincides with the geometry-recovery seed. Since seeds are supplied (Section 8.2), so is this coincidence (I2).

16. The retained gravitational-side sector (RS-1)

RS-1 tested five candidate sectors against two selectors.

- **The earned selector, given CARRIER:** the additivity ratio $A = R(0, 16)/R(0, 8)$ of the static effective resistance. A sector that carries an additive one-dimensional metric gives $A \approx 2$; the frozen window is $[1.8, 2.2]$.
- **The supplied selector, from the probe:** infrared boost compatibility, the ratio $v_g v_p / v^2$ of group and phase velocities, which tends to 1 for linear dispersion ($z = 1$, $\omega \propto k$) and fails for $z = 2$.

Candidate	$A = R(0, 16)/R(0, 8)$	$v_g v_p / v^2$	Status
Phonon	1.936 (additive)	0.999	survives both
Magnon ($z = 2$)	1.936 (additive)	3.993	eliminated by the supplied layer
Free fermions	1.020 (local density saturates)	0.995	eliminated by the earned layer (local access)
Flexural chain ($z = 2$)	3.744 (size-dependent)	3.993	eliminated by both
Gapped control	1.019 (screened)	0.999	eliminated by the earned layer

What survives is a class, relative to the tested candidates and in the infrared only:

gapless, $z = 1$, locally accessible Goldstone-like.

“Goldstone-like” means gapless, with static response $\propto 1/q^2$, reached through a local field [†Gol61].

This is a class, not a unique ontology, and the final (magnon) cut is made by the supplied probe structure. The gauge structure permits full-stress coupling but does not force it.

One frozen gate remains RED. The flexural gate predicted cubic growth, $A > 4$, and measured 3.744. The labeled diagnostic shows $R \sim Nr^2$ (quadratic and size-dependent) rather than cubic behaviour. The flexural sector is eliminated either way by the frozen window. The exact member of the class and its dispersion curvature belong to inventory item I11.

17. Matter–graviton coupling and the ω^7 result

This section gives the full record of the program’s most-cited gravitational number, including the parts that weaken it.

17.1 The setup and the mechanism

The instrument (coupling instrument v3, adjudicator-track, repair set 5f5395e; battery 20/20). A quantum harmonic chain of 256 sites, with lattice dispersion $\omega_q = \sqrt{\Omega^2 + 4 \sin^2(q/2)}$ (acoustic $\Omega = 0$; gapped control $\Omega = 1$), couples to weak-field TT gravitons in a periodic box (14,938 modes, two polarizations, $\omega = |k|$) only through its stress tensor, $L_{\text{int}} = -\frac{\kappa}{2} h_{ij} T^{ij}$. $J(\omega)$ is the golden-rule spectral density for a graviton of frequency ω to convert into a pair of phonons (q, q') , with energy and momentum conserved, at zero temperature and order κ^2 . The chain is the retained system and the gravitons are the bath.

The mechanism (GR-1, 30/31, d827a32). GR-1 built a fresh one-dimensional pair instrument sharing no code or data with v3, and wrote its predictions into its charter before any number existed. Every prediction landed:

- the kinetic-only and potential-only stress vertices each give a base exponent b (measured 4.002 and 4.004 in GR-1’s convention);
- a non-minimal admixture destroys the cancellation below (4.011);
- the full minimal-stress vertex gives $b + 4.004$: the order-2 tracelessness cancellation removes the leading term and leaves a curvature residual $-2\alpha q^4$;
- exactly linear dispersion makes the full vertex vanish identically ($< 10^{-14}$);
- all slopes are exactly invariant under rescaling the coupling by 16;
- the same-sign channel is kinematically closed at zero net momentum, and the gapped counterfactual is exactly empty below 2Ω .

In the v3 convention the base exponent is 3, so the channel’s exponent is $3 + 4 = 7$, and its memory kernel decays as t^{-8} . GR-1 grades the absolute value “consistent by mechanism at increment level, never one-formula derived”.

17.2 The measured values and the sealed ledger

Measured by v3:

- continuum exponent 7.0077 (fit window $\omega \in [0.05, 0.35]$);
- box exponent 7.018 (window $[0.15, 1.2]$);
- coefficient ratio 1.0005 at $\omega = 0.06$;
- kinetic-only 3.003, potential-only 3.006, exactly linear dispersion $\sim 10^{-34}$;
- the instrument’s own tidal control, +2.031 (predicted +2).

The sealed ledger. An analytic ledger (T2_THEOREM_GATE_AND_PREREGISTRATION_01.md) was committed at d2da3a5 on 24 September 2026 at 18:37 (UTC−5). It was written by a separate AI session (the “adjudicator”) without access to the rebuilt instrument’s output. Its exponent gate G5 is a local log-slope of 8.00 ± 0.05 over $\omega \in [0.02, 0.32]$, which is 7.00 ± 0.05 in the propagator convention.

What the comparison supports, and what it does not:

- *The exponent class matches.* 7.008 lies within the ledger’s 7.00 ± 0.05 , although on a different fit window. The frequently quoted “ 7 ± 0.15 ” is the instrument’s own tolerance, not the ledger’s.
- *The instrument’s pre-registration was committed with its result.* Its window, tolerance, and coefficient check are written in its docstring, which was first committed together with its result at 5f5395e. No earlier commit holds them.
- *The coefficient ratio is measured against the instrument’s own asymptotic formula*, not against the ledger’s sealed constants. The instrument labels this check “sealed G6”, but the mapping onto the ledger’s G6 constants is not implemented.
- *Several sealed gates have no implementation in v3:* the ledger’s synthetic pipeline test and normalization map (G1), its own fit window (G5), a parity fit (G7), the gapped-step coefficient (G8), a broadening stability check (G9), and its +4 tidal control. Only the instrument’s own +2 tidal variant was run.
- *Blindness is limited.* Two-sided blindness is attested only for the first rebuilt run (18:49 local time, timestamp-attested, not commit-sealed), which measured $\omega^{5.03}$ in its own convention. An adjudication committed after that run (ebb3a42) mapped the ledger’s exponent onto that convention as $8 - 2 - 1 = 5$. The v3 instrument was then built in the adjudicator’s tree, with pre-registrations mapped from the ledger, so v3 was not blind to it.
- *Both sides were AI-operated roles under the owner’s direction*, run as separate processes in separate trees. “Independent” here means separate processes and sealed commits, not external review. Matching one exponent class from a small discrete set is weak evidence.

The “Cherenkov no-go” episode. A rebuilt run at 19:13 reported $\omega^{2.38}$ and the verdict that the acoustic two-phonon channel is closed (“Cherenkov no-go”). An adjudication committed the same evening (1e09d30) refuted it: the no-go inequality holds only for same-sign pairs, 16 of 16 exact interior on-shell roots exist, and the run lacked the golden-rule $1/d\omega$ normalization; a removed kinetic term in the vertex was found later. The acoustic channel is **open**. The 18:49 and 19:13 run outputs were never committed.

17.3 What the result is

ω^7 is within-class occupancy evidence, not a universal GRUT prediction.

“Occupancy evidence” means that the ω^7 exponent class is inhabited by at least one explicitly constructed, consistently coupled microscopic sector. It is evidence that the class exists, not evidence that gravity’s actual dissipative response lies in it. Absolute exponents are never compared across conventions; GR-1’s own large-time slope, -8.975 , is reported only, because the sealed t^{-8} class lives in the v3 convention. GR-1 also shows the discretized ω^7 bath inside the cone, with its vacuum exactly on the floor: gravity inherits $\hbar$ as located; it does not generate it.

The Class-4 gate. The program’s criterion for a *forcing* derivation is that a claimed derivation of a supplied input counts only if it strictly reduces the total supplied content, without relocating, retyping, or restricting scope. For the exponent, GR-1 froze the rule: the gate passes only if no input beyond locality and symmetry is load-bearing. Measured:

- $\kappa \sim 1/M_{\text{Pl}}$ is **discharged** as amplitude-only: the slope is exactly invariant under rescaling it. This is not a derivation of Newton’s constant.
- the minimal-stress coupling is **load-bearing**: the non-minimal counterfactual loses the +4;
- the retained-sector structure is **load-bearing**: a gapless sector is a precondition for any low-frequency response.

The Class-4 gate remains open. Its residual obstruction is exact: the exponent is selected by counting *given* the coupling class, and the coupling class is supplied (I1–I6). Class-4 has never been passed.

18. The TT graviton-response sector (TT-1; 9/10)

Definitions and scope. The stress tensor admits an improvement term $\xi(\eta^{ij}k^2 - k^i k^j)f$, where ξ is its coefficient ($\xi = 0$ is the minimal coupling) and f runs over a tower of higher terms. The transverse-traceless projector about the graviton direction $\hat{k}$ is $\Lambda(X) = PXP - \frac{1}{2}P \text{tr}(PX)$, with $P = \mathbb{1} - \hat{k}\hat{k}$. TT-1 treats tree-level emission of one on-shell graviton ($\Omega = \omega_1 + \omega_2 = |k|$, $c = 1$) by a pair of retained-sector quanta with momenta p_1, p_2 , one isotropic dispersion per sector, no loops, and computes no rate exponent.

18.1 ξ is irrelevant on the physical TT channel

Every improvement structure has a TT image at most 2×10^{-15} of the canonical vertex. The reason is structural [IDENTITY]:

- $\Lambda(kk) = 0$;
- $\Lambda(\mathbb{1}) = 0$;
- $k^2 = 0$ on shell.

This is textbook transversality; its value here is classification. ξ remains visible off shell and in the trace, so CP-1’s finding for strain probes stands.

18.2 Tensor structure

The TT tensor structure is forced within the tested class (rank 1 at every point, $\lambda_2/\lambda_1 \sim 1.8 \times 10^{-16}$). So is the leading infrared coupling: the higher-derivative form factor is suppressed as p^2 , exactly $\times 4.000$ when p doubles. Higher form factors are constrained but not unique.

18.3 The rank-count red gate

The frozen prediction was rank 4; the measured rank is 3. An exact on-shell identity in the linear sector explains the difference. With $a = |p_1|$, $b = |p_2|$, $\omega_i = v|p_i|$, and $(\omega_1 + \omega_2)^2 = |p_1 + p_2|^2$,

$$p_1 \cdot p_2 = \frac{v^2 - 1}{2}(a^2 + b^2) + \omega_1 \omega_2.$$

Curved dispersion restores rank 4. The frozen gate nevertheless **remains RED**. A plausible explanation does not reclassify a failed gate.

18.4 Channel existence

Retained-sector dispersion	Channel
Exactly $v = c$, linear	Dead: the kinematics are collinear and the vertex vanishes identically
$v < c$, linear	Open (minimum vertex 0.032)
$v = c$ with subluminal curvature	Open , in proportion to the curvature (ratio 1.9975)
Superluminal curvature	Closed: no on-shell solution

With per-sector $v = c$ units, the channel exists only through dispersion curvature. That curvature is microscopic content the program has not earned (I11), and the comparison of the sector's light cone with the probe's is supplied (I6).

19. The 3D gravity red gate (GR-1)

GR-1 checked that the dimension entering the gravitational pair counting equals the dimension recovered spectrally in Section 9. The log-slope of the pair count should equal $2d$.

- In one dimension the gate passed: 2.222, inside 2 ± 0.4 , on the chain where G-2 recovered $\hat{d} = 1.000$.
- In three dimensions the frozen gate **failed and remains RED**: 5.362 against 6 ± 0.6 , a miss of 0.04 below the window.

A labeled diagnostic with the same frozen window on 13^3 , 21^3 , and 31^3 lattices gives $5.362 \rightarrow 5.493 \rightarrow 5.662$, rising monotonically toward 6; on 13^3 the lowest mode sits at 24% of the window edge. This suggests a refinement for a future instrument whose rule must be frozen in advance. It is not a repaired result: dimension consistency is certified in 1D and indicated, not gate-certified, in 3D.

20. The formal theory datum (Formalizations 01 and 02)

A GRUT datum is

$$\mathfrak{D} = (M, \mathfrak{a}, \varrho, \iota),$$

where:

- M is the microscopic model in an admitted class;
- $\mathfrak{a}$ is the access assignment (seed, closure, boundary events);
- ϱ is a **state** on M 's algebra (the formalizations write it ρ ; this record writes ϱ to keep ρ for the spectral measure);
- ι is the assignment of the conditional inventory (Section 21).

The **core datum** is $(M, \mathfrak{a}, \varrho)$. Two core data are **representation-equivalent**, $\sim_{\text{rep}}$, when an isomorphism of the underlying algebras intertwines dynamics, state, and seed. The record's controls verify that every core functional is constant on these classes (Givens scrambles, exact Lanczos re-representation, congruence checks). The core object is

$$\text{Core} : \mathbb{D}_{\text{core}} / \sim_{\text{rep}} \rightarrow (\mathcal{C}, \mathcal{I}, \mathcal{I}|_{\mathfrak{a}}, \mathcal{G}).$$

It satisfies seven compatibility relations:

Relation	Content	Grade
R1 kernel downstream	the kernel depends on model and access; exact spectral form; realization dimension = number of distinct eliminated modes	[DERIVED-IN-CLASS]
R2 admissibility	hierarchy positivity $\iff$ state positivity; Gaussian face in the cone, vacuum on the floor	[IDENTITY] + [DERIVED-IN-CLASS]
R3 interface completeness	for declared access, in class, the accessible hierarchy tracks every physical difference	[DERIVED-IN-CLASS]
R4 counting	sector $\mapsto$ exponent class by locality and symmetry; per branch, masked at eigenvalue level (gate red)	[DERIVED-IN-CLASS]
R5 geometry from data	geometry is a function of accessible influence data; selection graded by access class	[DERIVED-IN-CLASS]
R6 unit-change coherence	dimensionless accessible data discriminate unit changes; co-stretch $\equiv$ unit change	[IDENTITY]
R7 cross-fork datum sharing	the same data pushed through several maps give consistent outputs	checked property of the record

One theory on the core. The statement that GRUT is one theory on the core is a **checked property of the assembled record, not a theorem** about all admitted models. Upgrading it would mean proving R1–R6 as class-level theorems.

Domain extension. Following C1-a (Formalization 01 Amendment 02; equivalently Formalization 02 Amendment 01), $\mathbb{D}_{\text{core}}$ now covers finite stepped nonstationary models, in class, as well as stationary ones:

$$\mathbb{D}_{\text{core}}^{\text{stationary}} \longrightarrow \mathbb{D}_{\text{core}}^{\text{stationary}} \cup \mathbb{D}_{\text{core}}^{\text{stepped nonstationary}}.$$

This is the program’s first actual domain extension.

The conditional branch as an indexed family. The inventory space is $\mathbb{I} = \prod_{k=1}^{11} \mathbb{I}_k$. For each assignment ι , the fiber $\mathfrak{T}_\iota$ is the core object extended by the conditional maps (conservation, universality, sector selection, unit change, gravity) evaluated at ι . The formalizations call the resulting family a fibration,

$$\pi : \mathfrak{T} \rightarrow \mathbb{I}_{\text{adm}}, \quad \pi^{-1}(\iota) = \mathfrak{T}_\iota;$$

the word is used only for this indexed family, and no topological structure is claimed. The campaign did two things to it, each demonstrated in class by a chartered fork:

1. **It carved the base.** Six earned constraints carve the admissible region. Some remove assignments; the rest partition what survives:
 - a massless probe forces conservation (non-conserved couplings removed);
 - the cone cuts the classical carrier;
 - the sector class is reduced;
 - channel existence is partitioned by class;
 - universality is carved under exchange (non-universal clocks removed; decoupled assignments remain admissible);
 - the co-stretch dichotomy is real (both branches remain admissible; earned structure detects the difference but forbids neither).

2. **It refuted candidate sections.** A *section* over a coordinate is an earned map that would supply that coordinate from the core datum. Every attempted section family failed (P-6, CA-1, CC-1, U-1, SX-1, TT-1, P-2, and the prior Born-rule record).

The fibers were not shown to collapse. The program mapped where they remain.

The termination points of the typed maps, where a map stops at a supplied input or a seam, are part of the theory’s content, not omissions from it.

21. The current primitive-input inventory

For most structural assumptions above the influence layer, the campaign collapsed them into two stems: access, and probe structure. The quantum and boundary-data inputs are priced separately.

Stem A — access

- **I1. Access seed.** A subalgebra of operators (Section 8.2). Not derived; physical at boundary changes.
- **I2. CARRIER.** The identification “geometry-recovery seed = probe access seed”. Reduced to an access-seed coincidence (Section 15).

Stem B — probe structure

- **I3. The probe.** A massless, gauge- and Lorentz-redundant field with its coupling class. Supplied.
- **I4. Universal reach.** All sectors lie in one exchange-coupled component. **Supplied:** a probe coupled only through clocks mediates no exchange (U-1). Given I4, clock universality is derived in class; for sectors outside it, universality is irreducible.
- **I5. The co-stretch declaration (Sel-4x).** A rule for how constant probes act on intrinsic scales. Supplied.
- **I6. The cross-sector light cone.** A relation between sectors’ light cones. Supplied; this is seam S2.

Separately priced

- **I7. $\hbar$.** Located as the height of the fluctuation floor; not derived.
- **I8. Born measure and outcomes.** Not derived; the tested routes were negative.
- **I9. Noncommutativity.** Supplied.
- **I10. States and boundary data.** Amplitudes, states, the direction of the arrow of time, the low-entropy past, and (on an expanding background) a state choice, a temperature structure, and a stationary reduction.
- **I11. Microscopic sector content.** The support of ρ , τ_0 , the exact retained sector, its dispersion, and its curvature.

The inventory states the boundary of what the program has derived.

Certificate status (Formalization 02 §3.2). No coordinate yet holds a declared certificate.

- **I1 and I5** hold the *demonstration half* of an irreducibility certificate at stated scope: in-class exhibits of distinct values with identical accessible data.
- **I10** is the natural candidate for a domain datum.
- **I11** is the one coordinate with no certificate in either direction.

22. Seams, loopholes, and named obstructions

- **S1 — stationarity.** The finite stepped stratum is fully classified: the map and structure extend, and the packaging boundary is certified. Still open: C1-b (infinite volume) and C1-c (smooth modulation).
- **S2 — causal structure.** Spatial geometry is reconstructed in class, but no earned construction produces a common causal cone across sectors. Spatial and causal structure have not yet been derived from the same deeper data.
- **L1 — the free-sector loophole.** The retained free-phonon sector lies in the exception class of the clock-coupling ($O \propto H$) argument. This must be resolved before a self-consistent gravitational identification can be claimed.

Named obstructions (carried from Skeleton v02.1 §8 and Theory Paper Working Draft 02 §7):

- **Weinberg–Witten** [WW80] constrains massless spin-2 particles in theories with a Lorentz-covariant conserved stress tensor. It is carried as a **named-exit condition**: any eventual completion must state which hypothesis it relinquishes. Keeping the probe supplied (I3) is how this record currently stands relative to it.
- **Soft-graviton universality** [†Wei64, †Wei65]. Universal stress coupling is *imported* here as the minimal-stress postulate. That import is where the no-go pressure on emergent gravity concentrates.
- **Lorentz recovery** is open: a retarded kernel selects a frame, and the constructions here are non-relativistic lattices.
- **Energy bookkeeping** (the Bianchi identity) is open.

23. The preserved red register

Scope. This table lists the frozen gates that failed in the chartered forks on **master-w25bu9** (Appendix G). Its ten entries contain seventeen failed gates, because G-1 alone contributes eight. Section 23.1 lists the failed checks of the foundations record.

Fork / gate	Status	What failed
G-1 arrival-time geometry (eight gates)	RED	The arrival-time metric route fails in class: arrival-time functionals are not metric on the tested finite dispersive substrates at the tested scales
GR-1 3D dimension gate	RED	Measured 5.362 against a frozen 6 ± 0.6
P-6 degenerate-companion gate	RED	Measured exactly 0.0. Diagnostic: the companion operator annihilates the reachable even-parity block
P-3 H3 eigenvalue gate	RED	Measured +2.216 against a predicted +4. Diagnostic: eigenvalue masking; +3.999 on the cancelled branch
P-4 order-8 window gate	RED	Measured 67.8 with non-asymptotic couplings. Diagnostic: $325 \rightarrow 320 \rightarrow 276$, converging toward $2^8 = 256$
EQ-1 static discriminator	RED ; claim withdrawn	The “material mimic” was the geometric coupling in disguise (measured 2.7×10^{-10} against $> 10^{-3}$). The claim that static access separates what pair access cannot was withdrawn
RS-1 flexural gate	RED	Measured 3.744 against a predicted value above 4. Diagnostic: $R \sim Nr^2$
TT-1 rank count	RED	Measured rank 3 against a predicted 4. Diagnostic: an on-shell identity
C1-a same-lag drift	RED	Measured 0.0498 against a threshold of 0.1. Diagnostic: the peak normalization is modulation-independent

Fork / gate	Status	What failed
C1-a local-anchor family	RED	The midpoint anchor measured 0.0494 against a threshold of 0.05 for every member

Relation to C1-a2. The two C1-a reds are permanent. C1-a2 established a new result on the same question under a new pre-registration; it did **not** turn these reds green.

A red gate is not evidence against the whole theory. It is evidence against the specific claim or instrument that the gate represents.

23.1 Failed checks in the foundations record

The foundations record (`adjudicator-track`, Section 4) predates the charter protocol, and its failures are recorded here for completeness:

Instrument	Failed	What failed
irreversibility origin	1 of 4	the absorptive-spectrum sign check (Section 4.3)
minimal generative ontology	1 of 6	a derivative coupling outside the positive class; the check’s summary contradicts its data
gravity-bath spectral match	1 of 9	discrete-box kernel correlation 0.43 / 0.29 against 0.98
coarse-graining universality	5 of 7	five checks; several summaries contradict the recorded data
first coupling instrument	4 of 9	three exponent checks and the tidal control; the instrument was demoted and rebuilt (Section 17)
persistence origin P3, P4; spectrum E4a	3 checks	failed before the same-day instrument repair (Section 4.1; Appendix C.1)

24. The completion program

Completion means closing a finite set of typed problems. Each inventory coordinate must reach one of three certified statuses (Formalization 02, Definition 6):

1. **Derived section:** an earned derivation, certified by a chartered fork.
2. **Irreducible primitive:** an underdetermination demonstration at stated scope, followed by a declared axiom that carries the measured content.
3. **Domain datum:** explicit typing as boundary or state data within the theory’s domain, plus a check that the theory’s outputs are well defined given that datum.

GRUT is **closed as one theory** when the domain is enlarged across seams S1 and S2 (or each seam is proved a permanent boundary, with its new priced input named), and every coordinate carries one of the three statuses with its certificate. The goal is not to make every primitive disappear: “one theory” must not quietly mean “theory from nothing”. What is ruled out is an *uncertified* supplied input.

Problem	Success	Irreducibility route	Current status
C1 Stationarity seam	A nonstationary two-time kernel that reduces to the stationary one	Name the new priced structure the extension requires	Finite stepped stratum classified (C1-a, C1-a2). C1-b and C1-c open

Problem	Success	Irreducibility route	Current status
C2 Causal structure	A common cone from influence and access data under exchange	Show that relative cone data are not extractable at declared access	Open (seam S2)
C3 Access seed	A selector that survives the P-6 battery	A full-scope underdetermination result plus a declared axiom	Strong partial evidence; not closed
C4 Probe structure	Derive I3–I5 from declared alternatives	Classify the probe as an independent axis	Supplied
C5 Quantum scale and outcomes	A generative account of the floor; a Born derivation that survives an adversarial battery	A classification-grade negative result	Open
C6 Red gates	Separately frozen refinements, never reinterpretation	—	Open
C7 External QFT confrontation	Map the retained-sector class onto the T3 anchor	—	Protocol adopted; not executed

Listing a problem here does not imply that the program has a route to closing it.

Scope of C7. C7 tests the conditional branch, not the core. It compares the exponent class of the dissipative spectral density implied by the conditional branch with the T3 anchor’s, inside the anchor’s validity window. A mismatch there kills the gravitational identification, and the core survives as generic open-system results. A class match would be weak evidence, since matching one exponent class from a small discrete set is not unlikely by chance. A pass discharges no stem and no seam. How the matter-pair channel of Section 17 corresponds to the anchor’s pure-graviton loop is itself an open premise of C7. By a standing rider, any such comparison must state the stationarity regime it assumes (Section 5). Because C7’s premises depend on the retained-sector selection, loophole L1 must be addressed in them.

24.1 The T3 anchor

The anchor is a standard-QFT computation: the Wigner-time-resolved absorptive part of the one-loop pure-graviton self-energy on de Sitter space, through order H^6 , in the branch-cut class, with an ω^4 flat-space limit, a dissipative sign, and a validity window $\omega \gtrsim 3.4H$ (here H is the de Sitter Hubble rate). It is reported in *Wigner-time structure of one-loop graviton dissipation in de Sitter space* (D. Ryan Grover, version 1.0, 21 September 2026; frozen on branch `physics-final` at 310101f; present in the repository as `GRUT_de_Sitter_Absorptive_Response_v1.0/`). It is the previous version of this record under the same concept DOI.

- The object belongs to an established literature [TW96, TTW21]. What that paper certifies is its specific coefficient structure, not the object’s existence.
- Its verification is in-house; it has not been externally reviewed.
- It is **not** a GRUT claim, and it is not evidence for GRUT. Its ω^4 flat limit (in the self-energy convention) and the ω^7 of Section 17 (in the v3 convention) cannot be compared without the convention map that C7 must supply.

25. Falsifiability

GRUT currently has no GRUT-specific derived prediction (Section 29). The items below say how the declared layers or identifications could fail. The spectral-form and cone layers coincide with

standard open-system physics, so a violation there would challenge that physics, not a GRUT-specific hypothesis.

- **Spectral form.** The positive-measure form is a theorem for the declared passive class. It is falsified empirically only in a system that independent evidence places in that class; elsewhere a violation shows that the class does not apply. The record does not yet name any physical system as a claimed member of the class (a recorded gap, Skeleton v02.1 §7).
- **Influence cone.** A realizable state in the declared Gaussian class outside $J \geq 0$, $\nu \geq \hbar J/2$ would falsify the admissibility layer. Because the cone is state positivity (Section 6.4), such an observation would contradict the standard quantum mechanics of Gaussian open systems. Outside the class the cone does not apply.
- **Stationarity and packaging.** C1-a2 has a stated scope. Broader nonstationary classes can challenge whether the two-time datum is sufficient.
- **Gravity.** A mismatch in C7 would falsify the gravitational identification. It would not erase the continuum and influence results.
- **Cosmology.** The program’s one cosmological channel concerns the dark-energy equation of state $w(z)$.
 - *The statement.* A single passive relaxation mode cannot carry w across -1 (a generic result [Vik05]). The earlier Program Record release (7 September 2026) stated that every purely relaxational kernel in GRUT’s class stays on one side of $w = -1$, so an observed crossing would falsify the whole family, GRUT included.
 - *Its grade.* The program’s own ledgers hold this “no-crossing” statement at to-derive grade: whether GRUT’s vacuum-memory sector is such a mode depends on an unresolved link of its derivation chain, and the exclusion is shared by the whole passive class, not GRUT-specific.
 - *The sealed threshold.* A falsification threshold for DESI Data Release 3 was pre-registered and sealed on 18 August 2026 (provenance/prereg/PREREG_DESI_DR3_2026-08-18_v3.txt): a model-independent detection of $w(z) < -1$ at any redshift, at $\geq 5\sigma$, robust across the release’s headline data combinations, and published. The same file records that sustained non-crossing would confer no credit, and that the threshold is “this program’s commitment about its own model, not a test the cosmology literature would recognize as addressed to it”.
 - *Present data.* Published DESI analyses prefer evolving dark energy; the record cites about 3.1σ for DESI Data Release 2 with CMB data [DESI25]. The program’s own assessment is that the framework is not falsified by present data.

26. What GRUT does not claim

GRUT does **not** currently claim any of the following:

- a completed theory of everything;
- a derivation of the Standard Model (the matter link is silent);
- a derivation of $\hbar$, of the Born rule, or of noncommutativity;
- absolute geometry, or a unique microscopic ontology;
- a universal single-pole or stationary memory kernel;
- an absolute ω^7 prediction independent of sector and coupling assumptions;
- a completed gravitational identification, causal-cone derivation, or Class-4 result;
- resolution of every red gate;

- any confirmed novel quantitative prediction.

The claim this record supports is:

Within the tested model classes, GRUT has constructed and stress-tested a bottom-up hierarchy in which local microscopic dynamics generate continuum spectra, memory, irreversibility, influence structure, and access-relative geometry. It has also identified, systematically, where quantum, causal, access, probe, and gravitational structure cease to be derived and must currently be supplied or tested further.

27. Relation to GRUT-RAI

GRUT-RAI is not the physical theory. It is the computational and provenance infrastructure used to construct, test, audit, and preserve the theory. Its roles include:

- implementing models;
- running pre-registered gates;
- committing charters before runs;
- storing SHA-256-hashed results;
- running counterfactual batteries;
- detecting instrument defects (halts);
- preventing narrative status inflation;
- maintaining the dependency ledger.

Its intended workflow is deliberately adversarial. A cycle runs:

1. hypothesis;
2. counterexample;
3. charter, committed to git;
4. instrument and run;
5. verdict;
6. owner ruling;
7. dependency update.

Appendix D states what the git record does and does not establish about this ordering. RAI is a research instrument and a provenance system. It is not an authority that can make a physical claim true.

28. Research provenance and publication-source rule

Sources. The authoritative snapshot is `master-w25bu9` at `6abbf3164655c513b72dfc72f4a64013b061c489` (the source boundary), together with the foundations record on `adjudicator-track` at `90218f5`. The latter branch is not merged; its 51 files that are absent from the publication branch are archived byte-exact, with SHA-256 sums, in `archive/adjudicator-track_90218f5/`, so the deposit contains every source this record cites.

Branch chronology does not revise the theory. Claims from the RRP line (the Reality Requirements Program, recorded on `master` through `a2dcb02`, the fork point of `master-w25bu9`) enter only through source inspection, scope classification, and owner adjudication.

The consolidated record draws on:

- the foundations record (Section 4; Section 17’s coupling instrument, sealed ledger, and adjudications);
- the non-stationarity, kernel-transport, and clock-mismatch instruments;
- the chartered forks P-1 to P-6, D-1, S-1; G-1, G-2, GS-1; GR-1, CP-1, EQ-1, S4-1, FS-1, CC-1, U-1, RS-1, CA-1, SX-1, TT-1; C1-a and C1-a2;
- GRUT Working Theory 01; Theory Paper Working Drafts 01 and 02; Formalizations 01 and 02, with their dated amendments;
- the de Sitter absorptive-response paper, as the T3 anchor (GRUT_de_Sitter_Absorptive_Response_v1.0/).

Appendix G gives commits, dates, and battery scores. Owner rulings are recorded as files in the repository and on GitHub Issue #2; the issue thread itself is not part of a repository snapshot.

How to verify a result.

```
git clone https://github.com/ryangrivr/GRUT-RAI
cd GRUT-RAI && git checkout master-w25bu9
cp -r calc /tmp/grut_calc && cd /tmp/grut_calc
python3 c1a2_packaging.py
```

- The campaign instruments are pure Python 3 standard library and deterministic or fixed-seed. Each prints its battery (the command above prints “7/7 gated checks passed”) and writes its result file into the parent directory, so run them from a copy, as shown, to avoid overwriting the committed results.
- Each result embeds its wall-clock run time, so a re-run’s SHA-256 will differ from the one quoted in the verdict, while every recorded number should agree. For all 24 forks, the SHA-256 prefix quoted in the verdict matches the committed result file.
- The foundations instruments are in `archive/adjudicator-track_90218f5/calc/`. Six are pure standard library; the others need `numpy` (and two need `scipy`), with Python 3.12 recorded as the tested environment.

This manuscript and its release tooling were committed after the source boundary. They add no computation and change no status.

29. History, standing gradings, governance, and record integrity

Several earlier GRUT formulations were materially stronger than the present record supports. They are kept as history, not as claims.

29.1 Release history

Date	Release	Status now
Dec 2025 – May 2026	Closure Framework v1–v11; Phase I–III protocol deposits (e.g. 10.5281/zenodo.18008060); canonical builds (branch v1-retired); early GRUT-RAI software versions	Superseded
Apr – Jun 2026	GitHub releases v2.1.0 to v10.0, including “zero free parameters in the predictive core”, “ $R \approx 1.154$ with 0.28% precision”, and “candidate Theory of Everything”	Superseded. Nothing from these lineages is carried unless re-derived under the current discipline
18–19 Jun 2026	GRUT ToE v3.0.0, “The Corrected Physical Picture” (a 32σ CMB–ISW exclusion; a 689 Hz decoherence plateau)	Superseded; the 32σ figure recomputes to about 2.0σ

Date	Release	Status now
21 Jun 2026	<i>GRUT ToE v4.0 — The Emergence of Everything — Candidate Framework</i> (10.5281/zenodo.20783057, under concept DOI 10.5281/zenodo.19803663)	Withdrawn by its author (see the Notice at the front of this record)
7 Sep 2026	GRUT v5.0 <i>Program Record</i> : Books I–X working edition and the RAI audit infrastructure (under the same concept DOI)	Retired the theory-of-everything claim; superseded where it conflicts with this record
21 Sep 2026	<i>Wigner-time structure of one-loop graviton dissipation in de Sitter space</i> , v1.0 (the previous version of this record)	Stands as a standalone standard-QFT result and the C7 anchor (Section 24.1)
25 Sep 2026	The owner’s draft of this record (uploads/GRUT_Consolidated_Theory_2026-09-25_DRAFT_as_received.md)	Superseded by this edition
26 Sep 2026	This record (10.5281/zenodo.22983638)	Current

Version DOIs that the repository does not record are not given here; the Zenodo concept record lists every version.

29.2 Closure and reopening

- **Closure (23 September 2026).** The owner formally closed the program (GRUT_PROGRAM_CLOSURE_01.md). The closure recorded that the distinctive-theory claim failed, the theory-of-everything claim was retired, and no GRUT-specific prediction survived. It graded the founding hypothesis “unevaluated at its own claim point — neither proven nor disproven”. An owner amendment the same evening added that the closure “closes the construction, not the question that motivated it”.
- **Reopening (25 September 2026).** The owner reopened the program as a theory-development effort (GRUT_PROGRAM_REOPEN_01.md), with the object “exhibit the mathematics of the theory — show how it works — before any external claim is made”. The closure document is not retracted: it stands as the accurate record of the 23 September boundary. **Reopening is not rehabilitation:** every standing grading below carries over.

29.3 Standing gradings

- **Predictions.** The GRUT-specific prediction ledger stands at **zero derived predictions**. The signature audit found no admissible, parameter-free observable that distinguishes GRUT from standard physics (verdict EMPTY).
- **The distinctive-theory adjudication**, run twice on 4 and 5 September 2026 (a structural theory search with a blind hostile pass, and a resurrection audit), **failed** both times as scoped: the framework “fails to constitute a distinctive physical theory”. That is not the same as “the responsive-vacuum hypothesis is false”.
- **The founding bet**, a finite single-pole vacuum memory kernel, is **negated in its strongest form**: exact de Sitter free-field theory forces scale-free (power-law) memory at the computed order. The current theory builds on the computed structure; it does not re-argue the original ansatz.
- **The withdrawn June 2026 deposit** (10.5281/zenodo.20783057) stays withdrawn. Nothing in it is cited as a result.
- **Earlier no-go results stand at their recorded strengths** (see NO_GO_LEDGER.md):
 - *The α -bridge is settled-negative.* The hoped-for derivation of the tensor-response amplitude from the conformal-anomaly ratio $a/c = 1/3$ fails: the two anomalies live in different spin channels.

- $\mu = 4/3$ is excluded. The super-horizon enhancement of structure growth that GRUT’s conformal coefficient naively suggests fails a separate-universe consistency argument and is disfavoured empirically at a joint $\sim 4\sigma$ class (ISW cross-correlation $\sim 2.0\sigma$; DESI lensing $\sim 3.5\sigma$).
- General relativity is recovered with imports, not derived (Section 12).
- The Born rule is borrowed.

Demotions forced by the campaign:

- the single-pole kernel is not selected;
- τ_0 is not generated;
- the kernel sits downstream of partition, state, and access;
- (K, N) is not the whole interface;
- geometry is access-relative;
- stationarity is a regime;
- the gravity coupling is conditional on the supplied probe;
- ω^7 holds within its class;
- $\hbar$ is located, not derived;
- Born outcomes are unresolved.

29.4 Governance at publication

- **External review is deferred by owner direction.** Direction D-1 of the reopening ruling (25 September 2026) reads: “No dispatches, no solicitation of outside review, until the owner rules the mathematics stands.” It requires every artifact to carry the standing disclosure that no outside human has reviewed any of the work; this record does so. D-1 is not recorded as lifted. Publishing this record is an owner act, and it does not solicit review.
- **Reviewer vocabulary in the repository.** Where repository files speak of “external”, “specialist”, “referee”, “hostile”, “blind”, or “independent” reviewers, these were AI sessions run by the owner. No transmission to any outside human is logged at any date (`docs/WHERE_I_T_STOPS.md`); older wording to the contrary (for example in `STATE.md`) is superseded by that finding.
- **The signed termination condition.** A termination condition for in-house GRUT physics (version 4) was signed on 10 August 2026 and is in force. In-house calculation stops at the earliest of 31 December 2026 or its other stated conditions, after which a deposit reports each of five named channels in one line. No stop condition has fired, and this record is not that deposit. The reopening ruling states that it cannot discharge or amend the signed condition; the owner’s reconciliation of the two is owed.
- **The sealed DESI DR3 threshold** (Section 25) is in force. Whether it is the frozen crossing threshold that the termination condition’s cosmological channel requires has not been adjudicated.

29.5 Record integrity

- **Five narrative-versus-run mismatches** are documented in Skeleton v02.1 §10, cases in which a verdict’s prose asserted what its checks had not shown:
 1. a kernel-exponent mismatch, t^{-d} against $t^{-d/2}$ (continuum origin; repaired);
 2. a t^{-3} against t^{-4} inconsistency (spectral match; corrected in the script, while the committed result still carries t^{-3});
 3. a coupling verdict describing a run that did not happen (found by adjudication);

4. persistence and representation prose asserting a theorem whose checks had failed (P3, P4, E4a; found by two AI record-verifiers);
5. a kernel tail computed from a refuted pre-registration (repaired).

The systemic fix is that the coupling instrument now assembles every verdict string from measured variables.

- **The asymmetric error budget.** An audit of 4 September 2026 found that negative results had faced none of the checks applied to positive ones (RAI_GRUT_RESURRECTION.md). The codified response is the N1–N10 negative-control standard, including displaced-gate mutation testing. It is graded only partially repaired, “a detector, not a certifier”, and only N4 and N5 have committed definitions. The red-gate discipline complements it by keeping failed gates from being softened by prose. The “Cherenkov no-go” of Section 17, accepted and refuted the same evening, is an example of the asymmetry.
- **The foundations repair.** The same-day repair of 24 September 2026 regenerated the failing results in place at the branch head. That departs from the program’s earlier rule that a contaminated instrument is not patched; the originals survive in git history (Appendix C.1).
- **Found in preparing this edition** and disclosed in Sections 4 and 17: checks asserted in code rather than computed, a possible sign-convention defect, summaries that contradict their data, and sealed gates without implementation.

These are part of the record.

30. The best current conceptual picture

The program’s working picture is a heuristic, not a result. It reads, in the tested classes:

locality → hidden modes → continuum spectrum → memory → influence hierarchy

influence hierarchy + access → recoverable spectral structure → geometry

and, conditionally,

influence + access + geometry + probe structure → effective sector → gravitational-side response

Observable physical structure can be understood as a hierarchy of constraints generated by local dynamics and revealed through access, with memory, dissipation, spectral structure, and geometry emerging at successive coarse-grained levels.

The arrows differ in status. The first line is derived in class or family-fact; the second is derived in class and access-relative; the third is conditional on the supplied inputs I1–I6. The gravitational interpretation is a further identification, graded hypothesis. It is not the definition of the core.

30.1 Claim hygiene for citation and reuse

- **A constructive mechanism is not ontological uniqueness.**
- **An in-class derivation is not Class-4 forcing.** Fixed sector, coupling, probe, access, or state assumptions remain dependencies of the result.
- **Interface completeness is scoped** to the declared model and access class.

- **Geometry recovery is access-relative.**
- **The gravity branch is conditional.** Its classes and exponents are not universal predictions.
- **Red means red.** An explanation does not change a status.
- **Standard mathematics stays standard.** Where Section 2.1 or Section 31 marks an ingredient as standard, the record claims its in-class verification and its place in the map, not its discovery.
- **Provenance machinery is not physical evidence.**

31. Theory ledger at this source boundary

“Kind” separates what the program re-demonstrated from what it found: **S** = a standard ingredient, re-demonstrated in class; **P** = a program-specific in-class finding, not checked against the literature for priority; **—** = a supplied, open, or failed item.

Layer	Present status	Kind
Local microscopic dynamics	Constructive classes (Section 3.3); no unique ontology	—
Exact finite reduction (memory)	DERIVED-IN-CLASS	S
Continuum emergence	FAMILY-FACT (tested infinite-volume families; the finite reduction is DERIVED-IN-CLASS)	S
Irreversibility (existence)	FAMILY-FACT , as effective coarse-grained response; its direction is SUPPLIED	S
Positive spectral measure (class (a))	DERIVED-IN-CLASS	S
Recovery of ρ from the kernel	CONSTRAINED ; unique only for light tails	S
Spectral content of ρ (support, τ_0)	SUPPLIED (I11)	—
Nonstationary extension of the core map	DERIVED-IN-CLASS (finite, stepped)	P
Stationary packaging off stationarity	Certified not to package (finite, stepped, at the stated scope)	P
Two-time packaging datum	Exhibited constructively ; minimality open	P
Gaussian influence cone	DERIVED-IN-CLASS ; borrowed-standard mathematics	S
$\hbar$	SUPPLIED (located)	—
Full influence hierarchy	Complete interface in class ; its positivity is an [IDENTITY]	S
Born rule	SUPPLIED / UNRESOLVED	—
Noncommutativity	SUPPLIED	—
Access relativity	DERIVED-IN-CLASS	P
Access seed	SUPPLIED ; seed-to-closure map non-injective	P
Medium versus relational	Representational at fixed influence data, in class	S
Geometry reconstruction	DERIVED-IN-CLASS , access-relative (partial spectral geometry)	S, P
Absolute geometry	UNRESOLVED	—
Unit change	Co-stretch $\equiv$ unit change [IDENTITY]; rigid stretch observable	S
CARRIER	SUPPLIED ; reduced to an access-seed coincidence	P
Probe structure	SUPPLIED	—
Universal reach	SUPPLIED	—
Clock universality	DERIVED-IN-CLASS under exchange, given reach and the supplied probe; irreducible for decoupled sectors	S

Layer	Present status	Kind
Retained sector	Class DERIVED-IN-CLASS , conditional on CARRIER and the supplied probe, relative to the tested candidates; flexural gate RED ; member SUPPLIED / UNDEFINED (I11)	P
ω^7	Within-class occupancy evidence; not absolute	P
TT tensor structure and leading IR coupling	DERIVED-IN-CLASS	S
TT higher form factors	CONSTRAINED ; rank-count gate RED	—
3D gravity consistency	RED	—
Gravity as the memory bath	HYPOTHESIS ; spectrally viable (8/9); not derived	—
Causal / light-cone structure	SUPPLIED (S2)	—
Class-4 gravity	OPEN ; never passed	—
External QFT confrontation (C7)	OPEN	—
Confirmed novel predictions	None	—

32. Present status in one statement

As of 26 September 2026, at the source boundary defined at the beginning of this document:

GRUT has earned a generative open-system core. In it, within the admitted model classes, strictly local microscopic dynamics produce continuum spectral structure, persistent memory, effective irreversible response, and an influence hierarchy that is complete as an interface within the declared model and access class. Sufficiently rich access reconstructs tested graph and substrate geometry from accessible influence functionals. The core construction extends constructively to finite stepped nonstationary dynamics, where a pre-registered certification at the stated scope (with its calibration disclosed) finds that no single stationary spectral measure or Δt -only kernel packages the response. A conditional chain from influence, access, and geometry to a gravitational-side response has been organized, within the tested scope, into an explicit inventory of named primitive inputs and unresolved seams. The theory remains incomplete in the full nonstationary limit, causal structure, access-seed selection, probe derivation, quantum scale and outcomes, the gravity red gates, and external gravitational confrontation. It has no confirmed novel quantitative prediction, and no outside human has reviewed it.

This is the strongest statement the recorded evidence supports at this source boundary.

Appendix A — Core mathematical statements

	Statement	Scope and grade	Kind
A.1	$\dot{q} = -K_{SS}q + \int_0^t k(t-s)q(s) ds + \eta$, with $k(\tau) = \sum_l u_l^2 e^{-\lambda_l \tau}$, $\lambda_l > 0$	Class (a); exact; [DERIVED-IN-CLASS]	S
A.2	$k(\tau) = \int e^{-\tau/\theta} d\rho(\theta)$, $d\rho \geq 0$	Class (a), stationary; [DERIVED-IN-CLASS] (form)	S
A.2'	$K_0(t) = \int \cos(\omega t) d\mu(\omega)$	Class (b), infinite-volume limit; [FAMILY-FACT]	S
A.3	$J(\omega) \geq 0$, $\nu(\omega) \geq \frac{\hbar}{2} J(\omega)$	Declared Gaussian class; [DERIVED-IN-CLASS], borrowed-standard	S

	Statement	Scope and grade	Kind
A.4	$k(t, s) = k(t - s)$	Only on stationary backgrounds; recovered exactly when time-translation invariance is restored	P
A.5	$k(t, s)$ is two-time, packaged by $[\{\lambda_e, u_e\}, \{C_{e'e}\}]$	Finite stepped class; packaging boundary certified at stated scope; minimality open	P
A.6	Influence-hierarchy positivity $\equiv$ state positivity	[IDENTITY]; holds off stationarity by measurement	S
A.7	$\mathfrak{D} = (M, \mathfrak{a}, \varrho, \iota)$, with ϱ a state	Definition	—
A.8	$\pi : \mathfrak{T} \rightarrow \mathbb{I}_{\text{adm}}$	Definition; the base is carved in class	—

Appendix B — Major counterexamples that shaped the theory

1. **Single-pole non-uniqueness.** Many passive finite-memory kernels satisfy the same generic principles, and de Sitter forces scale-free memory at the computed order.
2. **KMS non-selection.** KMS and FDT states are points inside the cone.
3. **Medium/relational degeneracy.** Non-isomorphic baths share the same influence data (D-1).
4. **Higher-cumulant obstruction.** Identical (K, N) can give different probe physics (P-3).
5. **Access-seed non-uniqueness.** Identical closures can hide physically different seeds (P-6).
6. **Single-site geometry collapse.** A 2D grid and a 1D chain are identical from one site (G-2).
7. **The topology horizon.** Rings are indistinguishable below circumference order (G-2).
8. **Arrival-time failure.** Time of flight is not metric on the tested dispersive substrates (G-1).
9. **Exponent-locus non-uniqueness.** A plane of couplings gives the same exponent (CP-1).
10. **A surviving non-conserved coupling.** The phonon mass modulation passes every earned constant-level selector (CC-1).
11. **The free-sector conservation tower.** A free ring has $2R$ local conserved charges within range R (FS-1).
12. **Carrier non-uniqueness.** Four sectors carry identical geometry (CA-1).
13. **Exact-light-cone death of the TT channel (TT-1).** For an exactly light-cone-matched linear sector ($v = c$), the on-shell locus is collinear and the TT vertex vanishes identically.
14. **C1-a’s normalization defect.** Peak normalization suppresses nonstationarity.
15. **C1-a2’s stationary specificity control.** A different stationary kernel leaves the nonstationarity statistic at numerical zero.
16. **The refuted “Cherenkov no-go.”** A same-evening adjudication found the acoustic channel open (Section 17).

These counterexamples are part of the theory’s evidentiary structure.

Appendix C — The red-register rule

No red gate changes status through prose, reinterpretation, denominator changes, or retrospective threshold adjustment.

A failed gate stays red permanently. Its question can be re-tested only by a separately chartered refinement or an independent experiment whose scope, threshold, and adjudication rule are frozen before evaluation; such a re-test establishes a new result, and the original gate stays red. C1-a and C1-a2 are the worked example: C1-a’s two reds are permanent, and C1-a2’s certification is a new result under a new pre-registration, with its calibration disclosure on the face of its charter.

C.1 Instrument repair versus gate failure

- **Halts.** In the chartered campaign, an instrument that breaches an analytic identity halts: “instrument bug, never physics”, and no verdict may be issued from that run. A halt is not a red, and the repaired run is a new run.
- **Gate failures.** A frozen gate that fails stays red. A labeled post-hoc diagnostic may be recorded beside it, but no gate is repaired and no re-run is tuned.
- **The foundations exception.** The one place in this record where failed checks were re-banked after repair is the foundations record’s same-day repair set of 24 September 2026 (5f5395e). P3 had recorded the closed-loop rate instead of the kernel; P4 had a variable-shadowing bug and degeneracy-prone sampling; E4a compared a density with a step; and the coupling instrument had three defects (Section 17). The repairs were authorized by the owner and were not separately chartered. The failing results were regenerated in place; the originals remain in git at `affdf52`. The failures are listed in Section 23.1, and readers should weigh results re-banked after same-day repair accordingly.

Appendix D — Methods: the research protocol and the role of AI agents

D.1 The intended cycle

1. State the claim.
2. Identify the minimal assumptions.
3. Enumerate counterexamples.
4. Commit a charter (question, thresholds, decision rule) to git.
5. Implement the instrument.
6. Execute, without post-hoc tuning.
7. Preserve the failures.
8. Classify the result.
9. Obtain the owner ruling.
10. Update the dependency graph, and only then the theory ledger.

D.2 What “pre-registered” means here, and what it does not establish

- For each of the 24 chartered forks of Appendix G, a charter file was committed alone, and the instrument, result, and verdict were then committed together in the next commit. Each verdict commit’s parent is its charter commit.
- The gap between the two commits ranged from 76 s (SX-1) to 676 s (G-1), with a median of about 3 minutes.
- Because each instrument is first committed together with its result, the git record shows that each charter preceded its recorded result. It does not by itself show that no instrument was drafted or trial-run before its charter was committed; that rests on the operator’s process. Git timestamps are set by the committer, and no third-party timestamping was used. No run logs are committed.
- All 24 forks were chartered, run, and adjudicated on 25 September 2026, between 03:42 and 22:37 UTC. The foundations record was produced on 24 September 2026 without separately committed charters.
- **Calibration disclosures.** C1-a2’s thresholds were set from C1-a’s labeled diagnostics, as its charter states. No general record states who set the numerical thresholds of the other charters.

D.3 The role of AI agents

- Charters, thresholds, instruments, verdicts, the formalizations, and this manuscript were drafted by AI agents under the owner’s direction; the campaign’s commits are authored Claude.
- Owner rulings were given by the owner and recorded by the agent “from the owner’s own words”, with any misstatement to be corrected by owner edit.
- On the foundations branch, “builder” and “adjudicator” are AI-operated roles under the owner’s direction, working in separated trees. “Independent” there means separate processes and sealed commits, not external review.
- AI agents supplied computation, code, analysis, counterexample construction, document synthesis, and audit. They supply no authority. The physical status of a claim comes from the recorded construction, the committed instrument and result, and the explicit scope.

D.4 Battery counting

A battery score is checks passed out of the total, as recorded in each fork’s verdict. Through SX-1 the total counts every battery line, including frozen readings and labeled diagnostics; TT-1, C1-a, and C1-a2 count gated checks only. On gated checks alone, for example, G-1 is 9/17, P-6 is 23/24, and P-1 is 4/4. Failure counts do not depend on the convention. No record explains the change of convention.

D.5 Author, funding, and competing interests

The author is D. Ryan Grover, an independent researcher, the program’s owner and sole human participant. No funding and no competing interests are declared in the record.

Appendix E — The current research frontier

Foundational:

- C1-b and C1-c (the remainder of S1);
- C2 (causal cone, S2);
- C3 (access seed);
- C4 (probe structure);
- C5 ($\hbar$, Born outcomes, noncommutativity).

Integrity: the ten red-register entries of Section 23 (seventeen failed gates in all, eight of them G-1), plus the foundations-record failures of Section 23.1. The two C1-a reds are permanent; their question is answered by C1-a2.

Identification:

- L1 (the free-sector loophole);
- the Class-4 gravitational coupling;
- C7 (the external QFT confrontation).

These answer different questions, and none may be silently substituted for another.

Appendix F — Publication note and citation

This document is the Zenodo-facing statement of the current GRUT theory. It should be cited together with the GRUT-RAI repository snapshot, and with the committed artifacts whenever a

claim needs numerical provenance.

Recommended citation: Grover, D. Ryan (2026). *GRUT — Grand Responsive Universe Theory: Consolidated Theory, Formal Architecture, Results, Limits, and Completion Program*. Consolidated Research Record 02 (public-record edition), 26 September 2026. Zenodo. <https://doi.org/10.5281/zenodo.22983638> (this version). All versions: <https://doi.org/10.5281/zenodo.19803663>.

Software and provenance archive: Grover, D. Ryan (2026). *GRUT-RAI: research and provenance infrastructure for the GRUT program*. Zenodo. All versions: <https://doi.org/10.5281/zenodo.18993689>. The release corresponding to this record carries its own version DOI, minted on deposit.

Repository: <https://github.com/ryangrivr/GRUT-RAI>, branch `master-w25bu9` (source boundary `6abbf316`), with the foundations record of branch `adjudicator-track` (`90218f5`) archived in `archive/adjudicator-track_90218f5/`.

Version label. “Consolidated Research Record 02” is the owner’s label. No document titled Consolidated Research Record 01 is deposited or held in the repository; the earlier in-repository consolidations are GRUT Working Theory 01 and Theory Paper Working Drafts 01 and 02.

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The Zenodo record should preserve the repository snapshot that corresponds to this publication. This consolidation does not replace the detailed computational artifacts.

Appendix G — Campaign index (for archival verification)

G.1 The chartered forks

All forks are on `master-w25bu9`. “Battery” is the score recorded in each fork’s verdict (Appendix D.4). “Gap” is the time from the charter commit to the verdict commit. All verdicts are dated 25 September 2026 (UTC).

Fork	Question	Verdict (recorded strength)	Battery	Charter → verdict	Gap (min)	Verdict (UTC)
P-1	Can a subsystem be derived from correlations?	Regime trichotomy (structured / symmetric / generic), in class	6/6	<code>3bba895</code> → <code>aa569ce</code>	4.4	03:47
D-1	Medium or relational, at fixed (K, N) ?	Distinguishable mathematically; representational at system-limited access, in class	13/13	<code>53ab11f</code> → <code>baa64dc</code>	4.2	04:03
P-2	What minimally constrains realizable (K, N) ?	Cone confirmed: exactly two inequalities, in class; borrowed-standard	16/16	<code>291c23a</code> → <code>80982c1</code>	2.3	04:38
S-1	What selects a sector?	Selection by class (counting); no tested extremum	18/18	<code>351a326</code> → <code>cd3f7e3</code>	2.8	04:46
P-3	Does the cone survive noncommutativity?	Survives; null as a new principle; (K, N) is a projection	24/25	<code>78b188e</code> → <code>b05c99f</code>	5.3	05:01
P-4	Is the hierarchy characterized?	Characterized and reducible; ladder established	21/22	<code>d0fc014</code> → <code>d24b1bb</code>	4.3	05:12
P-5	Does a principle hide in access?	No; the seed is irreducible	22/22	<code>4013a32</code> → <code>eedd0ca</code>	4.4	05:22

Fork	Question	Verdict (recorded strength)	Battery	Charter → verdict	Gap (min)	Verdict (UTC)
P-6	Is the seed dynamically selectable?	No: non-derived input	38/39	179c108 → ee21a45	5.6	05:45
G-1	Arrival-time geometry	Partial; eight reds	26/34	f9b043b → aba55f1	11.3	06:12
G-2	Spectral geometry	Partial spectral geometry; single-site underdetermined	35/35	c83e9d7 → d612e2c	5.9	06:26
GR-1	The gravitational influence point and Class-4	Admissible; selected by structure in class; Class-4 open	30/31	ece56e7 → d827a32	6.5	07:02
CP-1	Minimal-stress coupling	Splits by dimension; reduced to geometric coupling	39/39	d78f9a4 → 5f5bd77	2.6	11:06
EQ-1	Geometric-only coupling	Reduced to Sel-4	45/46	708a31b → 6e66e28	3.2	15:15
S4-1	Sel-4	Splits by part and class	28/28	e83b6f0 → b2a2892	2.8	15:27
FS-1	The free-sector exception	Structural; $O = H$ not selected	31/31	19fd85f → c0099da	2.4	15:51
CC-1	C_cons	Irreducible; reduced to the massless probe	29/29	86a839f → 17c4395	2.8	16:23
U-1	Clock universality	Class split: derived in class under exchange, given the supplied probe	23/23	f31ab7d → 8e3232b	2.0	16:39
RS-1	Retained sector	A class selected in class, conditionally	27/28	1f18205 → a83421f	1.8	18:31
CA-1	CARRIER	Irreducible; an access-seed coincidence	23/23	e67b1d5 → a1ad8f9	1.3	18:37
GS-1	Geometry selection	Splits by access	19/19	7837fb2 → 8698f05	1.9	18:45
SX-1	Sel-4x (length equivalence)	Irreducible; the co-stretch declaration	23/23	3857e64 → 42e94cc	1.3	19:01
TT-1	The physical TT channel and ξ	ξ irrelevant; tensor forced; existence splits by class	9/10	b4a2f33 → ff3266d	4.0	19:14
C1-a	The stationarity seam: the map ε	ε extends; continuity and structure hold; packaging not certified	10/12	fa6ac44 → 55a708d	6.7	22:02
C1-a2	The packaging boundary	Certified (finite stepped, in class)	7/7	3b139ab → 4deda60	1.9	22:37

The owner’s acceptance of C1-a2 (Formalization 02 Amendment 02) is commit 6abbf316, 26 September 2026, 02:54 UTC: the source boundary.

G.2 Other instruments and records cited

Instrument or record	Branch	Result	Battery	Commit(s)
Non-stationarity measurement	master-w25bu9	Two-time at order unity on the sampled grid ($R = 0.516$; drift 1.53)	7/7	charter eb3fc07; result 5ea380e; Correction 01 6fc5139
Kernel transport	master-w25bu9	No tested local rule transports beyond $\sim 0.25/H_0$	11/11	charter c30f18e; verdict 05226bf
Clock mismatch	master-w25bu9	Comparison underdetermined	23/23	charter b25d79f; verdict 65774e4

Instrument or record	Branch	Result	Battery	Commit(s)
Foundations instruments (Section 4.5)	adjudicator-track	See Section 4.5	various	first committed affdf52 ; repairs 5f5395e
Coupling instrument v3	adjudicator-track	derived_within_class_onshell_omega7_channel_open	20/20	5f5395e
Sealed analytic ledger (T2 theorem gate)	adjudicator-track	Exponent gate G5: 7.00 ± 0.05 (propagator convention)	—	d2da3a5
Coupling adjudications 01–03	adjudicator-track	03 refutes the “Cherenkov no-go”	—	36caf65 , ebb3a42 , 1e09d30
GRUT Skeleton v02.1	adjudicator-track	Foundations synthesis	—	90218f5
T3 anchor (de Sitter paper v1.0)	physics-final	Standard-QFT result	—	310101f

Appendix H — Glossary

- **Access** (ℳ). Which operators an observer or probe couples to (the *seed*), everything they generate under products and time evolution (the *closure*), and changes of the seed (*boundary events*). Section 3.1.
- **Adjudicator, builder**. AI-operated roles under the owner’s direction on the foundations branch: one builds instruments, the other derives analytic expectations and audits (Appendix D.3).
- **Battery**. The full set of checks run by one instrument (Appendix D.4).
- **Branch (exponent)**. One channel of a matrix-valued influence, with its own exponent class (Section 7).
- **CARRIER**. The premise that the sector the gravitational probe couples to is the sector whose static response carries the recovered geometry (Section 15; I2).
- **C_cons**. The requirement that the probe’s constant limit couples to a conserved local charge, $[O, H] = 0$ (Section 13.1).
- **Charter**. The pre-registration committed before a run: question, thresholds, and decision rule (Appendix D.2).
- **Class-4 gate**. The criterion for a forcing derivation: it counts only if it strictly reduces total supplied content; for the exponent, only if nothing beyond locality and symmetry is load-bearing. Never passed (Section 17.3).
- **Co-stretch**. A rescaling of every coupling, intrinsic scales included: a pure unit change (Section 10).
- **Completely monotone**. A kernel that is a positive mixture of decaying exponentials (Section 4.4).
- **Continuum map** ε . The core’s map from microscopic model to continuum structure (Sections 5, 20).
- **Domain datum**. A coordinate typed as boundary or state data, with a well-definedness check; one of the three closure statuses (Section 24).
- **Earned**. Established by the program’s own committed instruments at [DERIVED-IN-CLASS] strength or better; the opposite of supplied.
- **Fork**. One chartered investigation (for example P-2 or CA-1).
- **Gate**. A single pre-registered pass/fail test within a battery.
- **GeoInv**. An unpromoted candidate: a constant probe must not rewire a sector’s recovered hop geometry (Section 14).

- **Halt.** An instrument stop on an analytic-identity breach: “instrument bug, never physics”. “Halt-grade” identities are checks whose failure would halt the run (Appendix C.1).
- **In class.** Established only within a declared model class (Section 3.3); the class is part of the statement.
- **Influence data, influence hierarchy.** The response, noise, and higher multi-time correlations through which an environment acts on the retained system (Section 6).
- **Inventory (I1–I11).** The explicitly supplied inputs (Section 21).
- **Irreducible (fork verdict).** Within the tested battery, no earned structure selects the input. This is not the formal “irreducible primitive” status of Section 24, which needs a declared certificate.
- **Labeled diagnostic.** A post-hoc measurement, labeled as such, recorded beside a failed gate; it never repairs the gate.
- **LocPos.** A candidate criterion: every edge weight and pin is non-negative (Section 9.4).
- **Matched control.** A replay of the same data under a relabeling, or a recomputation of a prior result, used to show the instrument reads zero where it should.
- **Min-dominance.** At low frequency the lowest exponent among channels dominates (Section 7).
- **NULL-REDUNDANT; NULL-AS-NEW-PRINCIPLE.** A candidate principle that reduced to standard structure or to an existing identity (Section 1).
- **Occupancy evidence.** Evidence that an exponent class is inhabited by at least one consistently coupled microscopic sector (Section 17.3).
- **Owner ruling.** The author’s adjudication of how a result is admitted (Section 1).
- **Packaging.** Representing a two-time kernel by a single stationary spectral measure or a single function of $t - s$ (Section 5.2).
- **Prediction.** In this record, an empirical prediction about nature. Agreements between in-house analytic and numerical computations are called checks.
- **Probe.** The gravitational test field h_{ij} , modelled as a strain, a clock field, or a massless spin-2 field (Section 12); elsewhere a probe qubit used to read out coherence.
- **Realization dimension.** The minimal number of hidden modes that reproduces a kernel (its Hankel rank).
- **Red.** A failed pre-registered gate; permanent (Appendix C).
- **Retained sector.** The degrees of freedom kept after eliminating the environment; on the gravity side, the matter sector coupled to the probe.
- **RRP.** The Reality Requirements Program, a separate line on the **master** branch (Section 28).
- **S-level, S-local.** Using only data of the retained system S (Section 5.1); observables of S alone (Section 8.4).
- **Seam.** A place where the theory’s objects are not yet defined: S1 (stationarity), S2 (causal structure).
- **Section.** An earned map that would supply an inventory coordinate from the core datum (Section 20).
- **Sel-4, Sel-4t, Sel-4U, Sel-4x.** A constant probe acts on each sector as a pure change of units (the operational weak equivalence principle); its temporal part, universality part, and length part (Sections 12, 14, 10).
- **Stem.** A group of inventory inputs: A (access, I1–I2) and B (probe, I3–I6).
- **T3 anchor.** The standard-QFT de Sitter graviton self-energy calculation used by C7 (Section 24.1).
- **TT.** Transverse-traceless: the physical polarizations of a graviton.
- **Universal reach.** All sectors lie in one exchange-coupled component (I4).

- z . The dynamical exponent, $\omega \propto k^z$; $z = 1$ is linear (sound-like) dispersion.

Appendix I — Notation and symbols with more than one meaning

Symbol	Meaning in this record	Other meanings in the program’s files
$k(\tau), k(t, s)$	Memory kernel of Sections 4–5, decay-rate form, $\lambda_l > 0$	Written $K_R(t) = \sum (v_{1k})^2 e^{\lambda_k t}$, $\text{Re } \lambda_k < 0$, in earlier documents
$K(t), N(t)$	Feynman–Vernon dissipation and noise kernels (Section 6)	N is also a system size, and a realization dimension in the foundations files
$\mathbf{K}, K_{SS}, \mathbf{K}_{yy}, K_{AI}$	Stiffness matrix and its blocks	—
$K_0(t)$	Local kernel of a conservative lattice (Section 4.2)	—
$J(\omega), \nu(\omega)$	Dissipation and noise spectra; matrix-valued in Section 6.3	J_0 is a Bessel function (Section 4.2); J is a drive or a symplectic matrix in some instruments
$\rho(\theta), \mu(\omega)$	Spectral measure (class (a)); local density of states (class (b))	—
ϱ	A state (Section 20)	The formalizations write it ρ
s	Symmetric occupation (Section 6.1)	Low-frequency exponent, $J \propto \omega^s$ (Section 7)
ξ	Improvement coefficient (Sections 12, 18)	SX-1 writes the correlation length as $\xi^2 = k/p$; this record writes ℓ_c
H	Hamiltonian	H_0 : present Hubble rate (Section 5); H : de Sitter Hubble rate (Section 24.1); H3: an S-1 hypothesis label
R	Residual fraction (Section 5)	$R(0, r)$: effective resistance (Sections 15–16); R1–R7: relations (Section 20); R : charge range (Section 14)
κ	Gravitational coupling, $\sim 1/M_{\text{Pl}}$	κ_4 : fourth cumulant (Section 6.3)
C	Mixing matrix $V_{\text{new}}^T V_{\text{old}}$ (Section 5)	$C_{ab}(\omega)$: bath correlation spectrum; C_{cons} ; C1–C7; $\mathcal{C}$: continuum object
P-1 ... P-6	Forks (Sections 6–8)	C1-a2’s charter gates (renamed A2-1 ... A2-6 here); the foundations checks P1–P7
S-1; S1, S2	A fork; the two seams	S-level, S-local (Glossary)
D-1	A fork (Section 8.4)	Owner direction D-1, deferring external review (Section 29.4)
T3	The de Sitter anchor	A target label in the P-2 instrument

Units and conventions.

- $\hbar = k_B = 1$ in the P-2, P-3, CA-1, and GR-1 instruments; Section 6.1 restores $\hbar$ by declaration.
- CA-1’s classical noise, $\nu = (2T/\omega)J$, is twice the high-temperature limit of P-2’s thermal curve; this affects only CA-1’s finite-temperature statements, not the $T = 0$ value quoted in Section 15.
- Exponent conventions are never mixed: the v3 propagator convention (ω^7, t^{-8}), the sealed ledger’s task-literal convention (ω^8), GR-1’s own convention (base ≈ 4), and the T3 self-energy convention (ω^4 flat limit) are compared only through increments or through an explicit convention map.

References

Citation keys without a dagger are held in the program’s own source register. The register records a verification tier for each entry: most here were checked at metadata or abstract level (via Crossref, arXiv, or the publisher) on 23 September 2026, and some are noted as known only through secondary literature. **References marked † are supplied for this edition. Their bibliographic data have not been verified against the program’s source register.** No systematic literature search has been performed for this record.

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